

CAN FD Transceiver with Wake-up Pattern (WUP) Option

Features

- Supports CAN 2.0 and CAN with Flexible Data-Rate (CAN FD) Physical Layer Transceiver Requirements
- Optimized for CAN FD at 2, 5 and 8 Mbps Operation:
 - Maximum propagation delay: 120 ns
 - Loop delay symmetry: -10%/+10% (2 Mbps)
- MCP2542FD/4FD:
 - Wake-up on CAN activity, 3.6 μ s filter time
- MCP2542WFD/4WFD:
 - Wake-up on Pattern (WUP), as specified in ISO 11898-2:2016, 3.6 μ s activity filter time
- Implements ISO 11898-2:2016
- Qualified According to AEC-Q100 Rev. G
- Very Low Standby Current (4 μ A, typical)
- VIO Supply Pin to Interface Directly to CAN Controllers and Microcontrollers with 1.8V to 5V I/O
- CAN Bus Pins are Disconnected when Device is Unpowered:
 - An unpowered node or brown-out event will not load the CAN bus
 - Device is unpowered if VDD or VIO drop below its POR level
- Detection of Ground Fault:
 - Permanent dominant detection on TXD
 - Permanent dominant detection on bus
- Automatic Thermal Shutdown Protection
- Suitable for 12V and 24V Systems
- Meets or Exceeds Stringent Automotive Design Requirements, Including “*Hardware Requirements for LIN, CAN and FlexRay Interfaces in Automotive Applications*”, Version 1.3, May 2012:
 - Conducted emissions @ 2 Mbps with Common-Mode Choke (CMC)
 - Direct Power Injection (DPI) @ 2 Mbps with CMC
- Meets SAE J2962/2 “*Communication Transceiver Qualification Requirements - CAN*”:
 - Radiated emissions @ 2 Mbps without a CMC
- High Electrostatic Discharge (ESD) Protection on CANH and CANL, meeting IEC61000-4-2 up to ± 13 kV
- Temperature Ranges:
 - Extended (E): -40°C to +125°C
 - High (H): -40°C to +150°C

Description

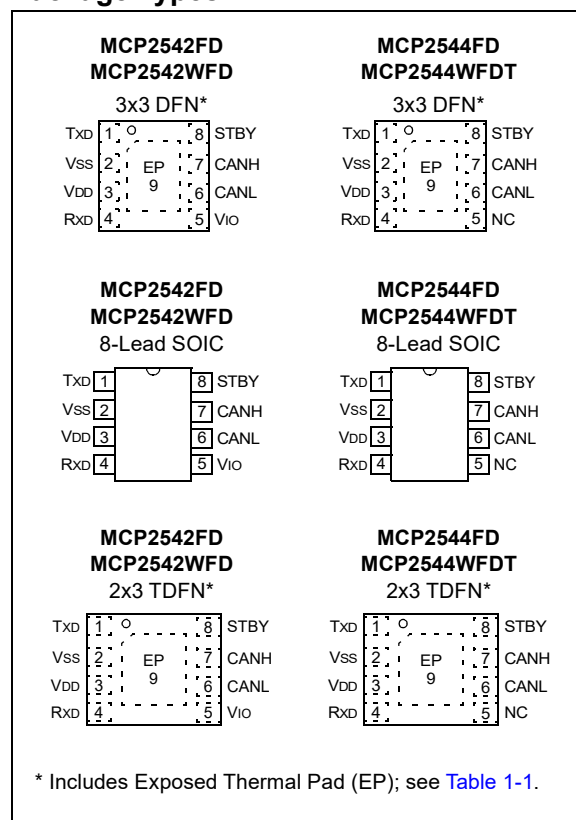
The MCP2542FD/4FD and MCP2542WFD/4WFD CAN transceiver family is designed for high-speed CAN FD applications of up to 8 Mbps communication speed. The maximum propagation delay was improved to support longer bus length.

The device meets the automotive requirements for CAN FD bit rates exceeding 2 Mbps, low quiescent current, Electromagnetic Compatibility (EMC) and Electrostatic Discharge (ESD).

Applications

CAN 2.0 and CAN FD networks in automotive, industrial, aerospace, medical and consumer applications.

Package Types



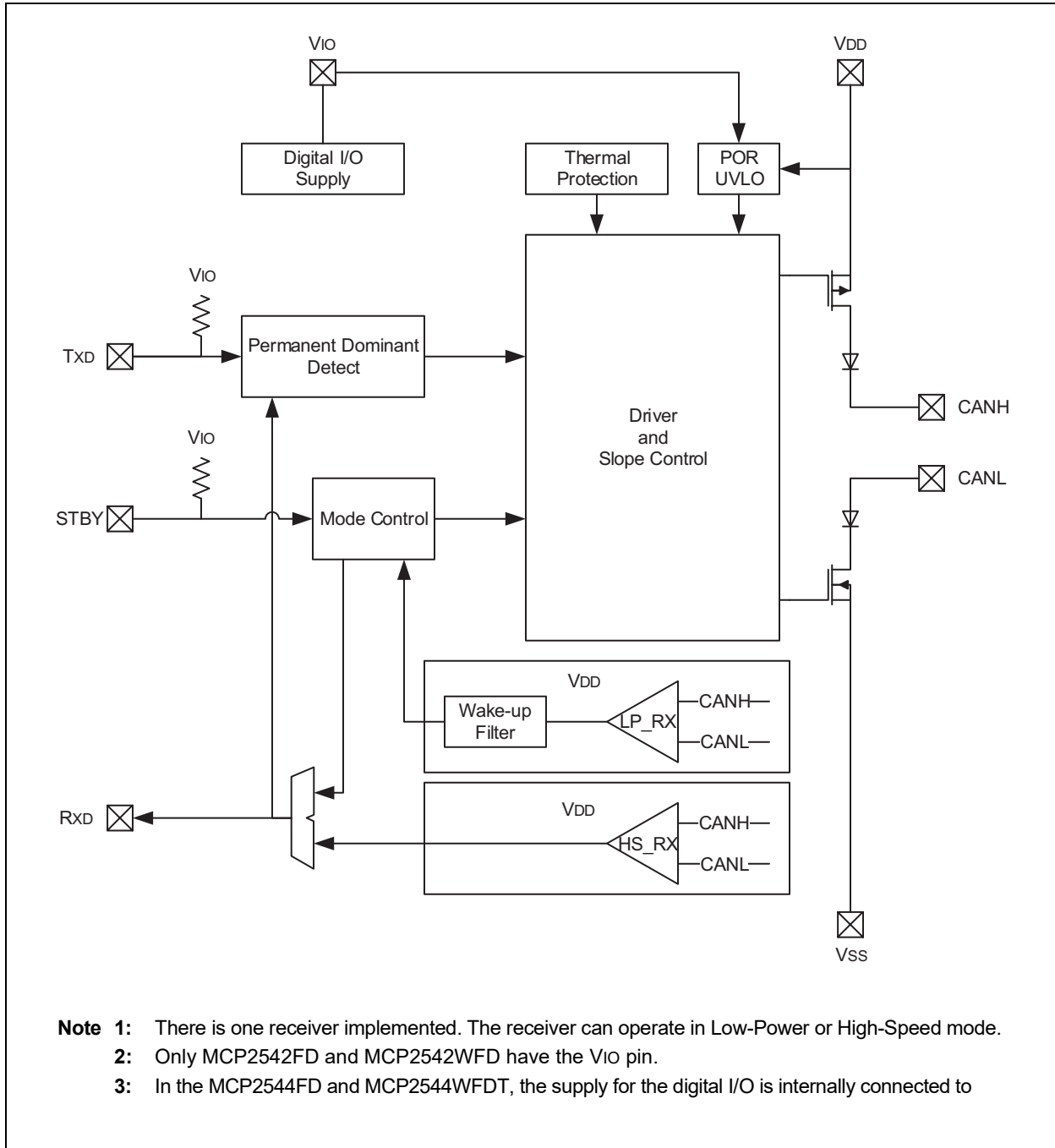
MCP2542FD/4FD, MCP2542WFD/4WFD

MCP2542FD/4FD, MCP2542WFD/4WFD Family Members

Device	Vio pin	WUP	Description
MCP2542FD	Yes	No	
MCP2544FD	No	No	Internal Level Shifter on Digital I/O Pins
MCP2542WFD	Yes	Yes	Wake-up on Pattern (see Section 1.6.5 “Remote Wake-up via CAN Bus (WUP)”)
MCP2544WFDT	No	Yes	Internal Level Shifter on Digital I/O Pins; Wake-up on Pattern

Note: For ordering information, see the [Product Identification System](#) section.

Block Diagram



MCP2542FD/4FD, MCP2542WFD/4WFD

1.0 DEVICE OVERVIEW

The MCP2542FD/4FD and MCP2542WFD/4WFD devices serve as the interface between a CAN protocol controller and the physical bus. The devices provide differential transmit and receive capability for the CAN protocol controller. The devices are fully compatible with ISO 11898-2:2016.

Excellent loop delay symmetry supports data rates up to 8 Mbps for CAN FD. The maximum propagation delay was improved to support longer bus length.

Typically, each node in a CAN system must have a device to convert the digital signals generated by a CAN controller to signals suitable for transmission over the bus cabling (differential output). It also provides a buffer between the CAN controller and the high-voltage spikes that can be generated on the CAN bus by outside sources.

The MCP2542FD/4FD wakes up on CAN activity (basic wake-up). The CAN activity filter time is 3.6 μ s maximum.

The MCP2542WFD/4WFD wakes up after receiving two consecutive Dominant states separated by a Recessive state: WUP. The minimum duration of each Dominant and Recessive state is t_{FILTER} . The complete WUP has to be detected within $t_{\text{WAKE(To)}}$.

1.1 Transmitter Function

The CAN bus has two states: Dominant and Recessive. A Dominant state occurs when the differential voltage between CANH and CANL is greater than $V_{\text{DIFF(D)(I)}}$. A Recessive state occurs when the differential voltage is less than $V_{\text{DIFF(R)(I)}}$. The Dominant and Recessive states correspond to the Low and High states of the TXD input pin, respectively. However, a Dominant state initiated by another CAN node will override a Recessive state on the CAN bus.

1.2 Receiver Function

In Normal mode, the RXD output pin reflects the differential bus voltage between CANH and CANL. The Low and High states of the RXD output pin correspond to the Dominant and Recessive states of the CAN bus, respectively.

1.3 Internal Protection

CANH and CANL are protected against battery short circuits and electrical transients that can occur on the CAN bus. This feature prevents destruction of the transmitter output stage during such a Fault condition.

The device is further protected from excessive current loading by thermal shutdown circuitry that disables the output drivers when the junction temperature exceeds a nominal limit of +175°C.

All other parts of the chip remain operational and the chip temperature is lowered due to the decreased power dissipation in the transmitter outputs. This protection is essential to protect against bus line short-circuit induced damage. Thermal protection is only active during Normal mode.

1.4 Permanent Dominant Detection

The MCP2542FD/4FD and MCP2542WFD/4WFD devices prevent two conditions:

- Permanent Dominant condition on TXD
- Permanent Dominant condition on the bus

In Normal mode, if the MCP2542FD/4FD and MCP2542WFD/4WFD devices detect an extended Low state on the TXD input, they will disable the CANH and CANL output drivers in order to prevent the corruption of data on the CAN bus. The drivers will remain disabled until TXD goes high. The high-speed receiver is active and data on the CAN bus are received on RXD.

In Standby mode, if the MCP2542FD/4FD and MCP2542WFD/4WFD devices detect an extended dominant condition on the bus, it will set the RXD pin to a Recessive state. This allows the attached controller to go to Low-Power mode until the dominant issue is corrected. RXD is latched high until a Recessive state is detected on the bus and the wake-up function is enabled again.

1.5 Power-on Reset (POR) and Undervoltage Detection

The MCP2542FD/4FD and MCP2542WFD/4WFD have POR detection on both supply pins: VDD and VIO. Typical POR thresholds to deassert the Reset are 1.2V and 3.0V for VIO and VDD, respectively.

When the device is powered on, CANH and CANL remain in a high-impedance state until VDD exceeds its undervoltage level. Once powered on, CANH and CANL will enter a high-impedance state if the voltage level at VDD drops below the undervoltage level, providing voltage brown-out protection during normal operation.

In Normal mode, the receiver output is forced to the Recessive state during an undervoltage condition on VDD. In Standby mode, the low-power receiver is designed to work down to 1.7V VIO. Therefore, the low-power receiver remains operational down to VPORL on VDD (MCP2544FD and MCP2544WFD). The MCP2542FD and MCP2542WFD transfer data to the RXD pin down to 1.7V on the VIO supply.

MCP2542FD/4FD, MCP2542WFD/4WFD

1.6 Mode Control

The main difference between the MCP2542FD/4FD and MCP2542WFD/4WFD is the wake-up method.

[Figure 1-1](#) shows the state diagram of the MCP2542FD/4FD. The devices wake up on CAN activity.

[Figure 1-2](#) shows the state diagram of the MCP2542WFD/4WFD. The devices wake up on a WUP.

1.6.1 UNPOWERED MODE (POR)

The MCP2542FD/4FD and MCP2542WFD/4WFD enter Unpowered mode under the following conditions:

- After powering up the device, or
- If VDD drops below VPORL, or
- If VIO drops below VPORL_VIO

In Unpowered mode, the CAN bus will be biased to ground using a high-impedance. The MCP2542FD/4FD and MCP2542WFD/4WFD are not able to communicate on the bus or detect a wake-up event.

1.6.2 WAKE MODE

The MCP2542FD/4FD and MCP2542WFD/4WFD devices transition from Unpowered mode to Wake mode when VDD and VIO are above their PORH levels. From Normal mode, the devices will also enter Wake mode if VDD is smaller than VUVL, or if the band gap output voltage is not within valid range. Additionally, the device will transition from Standby mode to Wake mode if STBY is pulled low.

In Wake mode, the CAN bus is biased to ground and RXD is always high.

1.6.3 NORMAL MODE

When VDD exceeds VUVH, the band gap is within valid range and TXD is high; the device transitions into Normal mode. During POR, when the microcontroller powers up, the TXD pin could be unintentionally pulled down by the microcontroller powering up. To avoid driving the bus during a POR of the microcontroller, the transceiver proceeds to Normal mode only after TXD is high.

In Normal mode, the driver block is operational and can drive the bus pins. The slopes of the output signals on CANH and CANL are optimized to reduce Electro-magnetic Emissions (EME). The CAN bus is biased to VDD/2.

The high-speed differential receiver is active.

1.6.4 STANDBY MODE

The device may be placed in Standby mode by applying a high level to the STBY pin. In Standby mode, the transmitter and the high-speed part of the receiver are switched off to minimize power consumption.

The low-power receiver and the wake-up block are enabled in order to monitor the bus for activity. The CAN bus is biased to ground.

The RXD pin remains high until a wake-up event has occurred.

The MCP2542FD/4FD uses basic wake-up: one dominant phase for a minimum time of tFILTER will wake up the device.

The MCP2542WFD/4WFD will only wake up if it detects a complete WUP. The WUP method is described in the next section.

After a wake-up event was detected, the CAN controller gets interrupted by a negative edge on the RXD pin.

The CAN controller must put the MCP2542FD/4FD and MCP2542WFD/4WFD back into Normal mode by deasserting the STBY pin in order to enable high-speed data communication.

The CAN bus wake-up function requires both supply voltages, VDD and VIO, to be in valid range.

1.6.5 REMOTE WAKE-UP VIA CAN BUS (WUP)

The MCP2542WFD/4WFD wakes up from Standby/Silent mode when a dedicated Wake-up Pattern (WUP) is detected on the CAN bus. The Wake-up Pattern is specified in ISO 11898-6 and ISO 11898-2:2016 (see [Figure 1-2](#) and [Figure 2-11](#)).

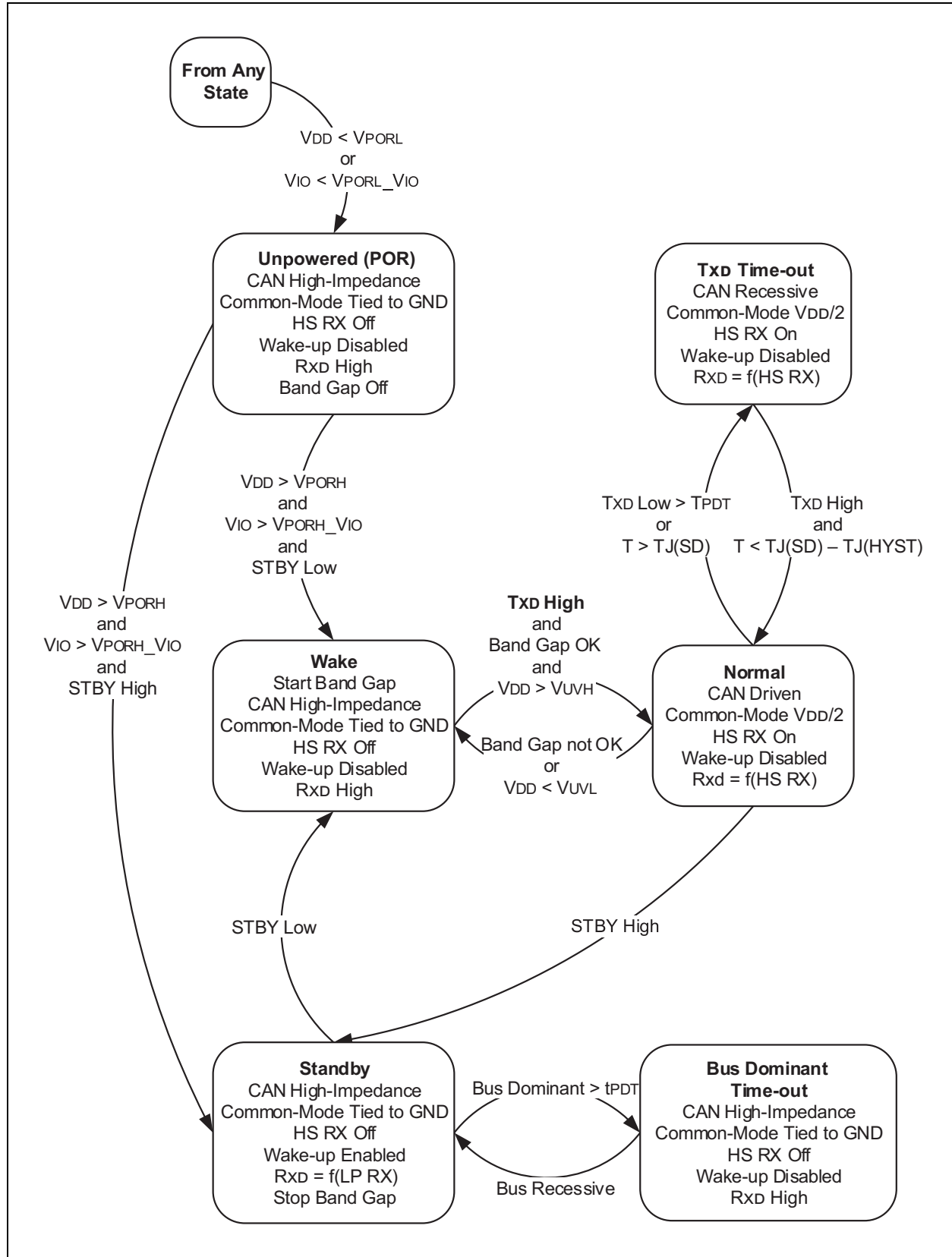
The Wake-up Pattern consists of three events:

- A dominant phase of at least tFILTER, followed by
- A recessive phase of at least tFILTER, followed by
- A dominant phase of at least tFILTER

The complete pattern must be received within tWAKE(TO). Otherwise, the internal wake-up logic is reset and the complete Wake-up Pattern must be retransmitted in order to trigger a wake-up event.

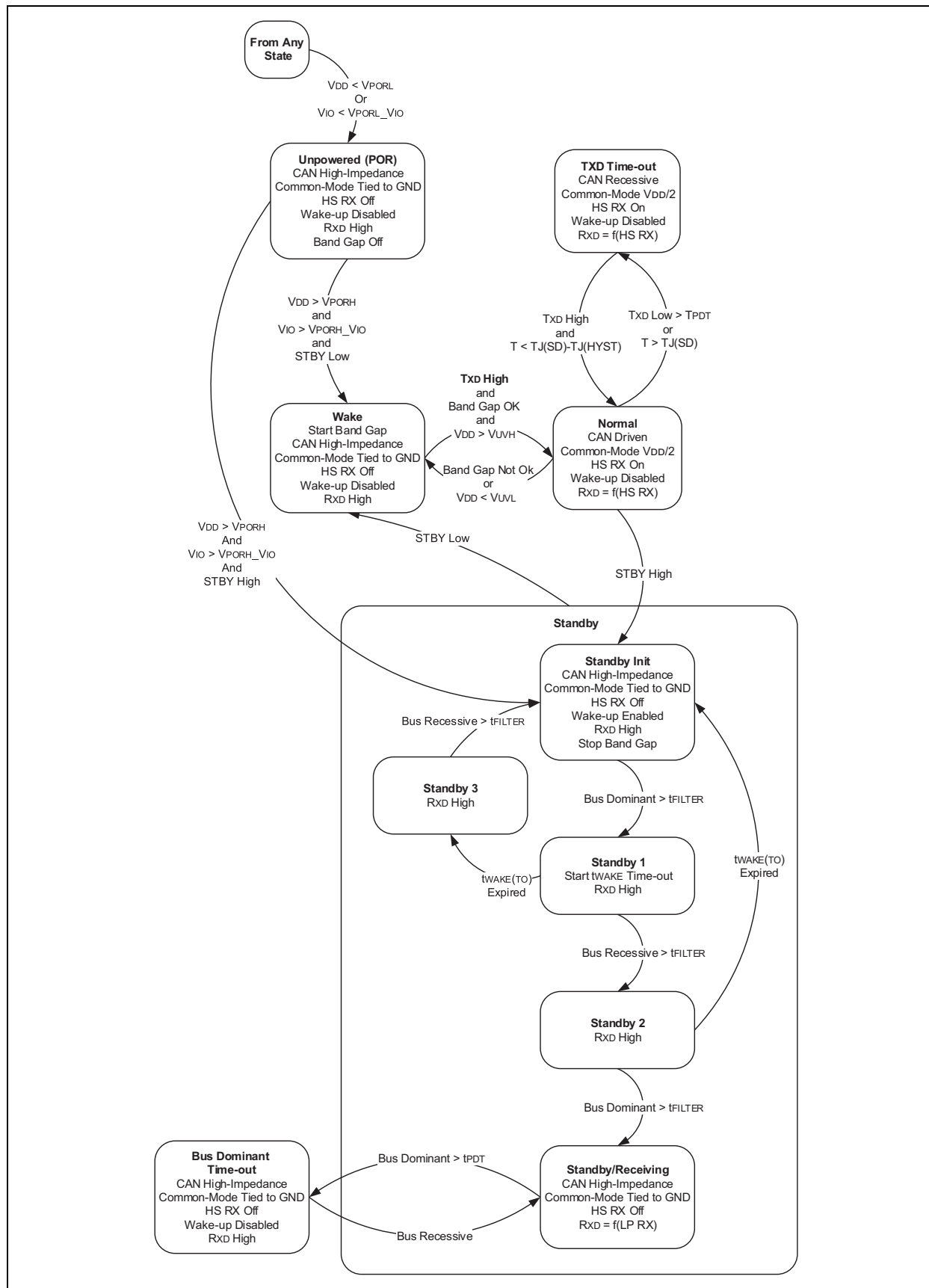
MCP2542FD/4FD, MCP2542WFD/4WFD

FIGURE 1-1: MCP2542FD/4FD STATE DIAGRAM: BASIC WAKE-UP



MCP2542FD/4FD, MCP2542WFD/4WFD

FIGURE 1-2: MCP2542WFD/4WFD STATE DIAGRAM: WAKE-UP PATTERN



MCP2542FD/4FD, MCP2542WFD/4WFD

1.7 Pin Descriptions

The description of the pins are listed in [Table 1-1](#).

TABLE 1-1: MCP2542/4FD AND MCP2542/4WFD PIN DESCRIPTIONS

MCP2542FD MCP2542WFD 3x3 DFN, 2x3 TDFN	MCP2542FD MCP2542WFD SOIC	MCP2544FD MCP2544WFDT 3x3 DFN, 2x3 TDFN	MCP2544FD MCP2544WFDT SOIC	Symbol	Pin Function
1	1	1	1	TxD	Transmit Data Input
2	2	2	2	VSS	Ground
3	3	3	3	VDD	Supply Voltage
4	4	4	4	RxD	Receive Data Output
—	—	5	5	NC	No Connect
5	5	—	—	VIO	Digital I/O Supply Pin
6	6	6	6	CANL	CAN Low-Level Voltage I/O
7	7	7	7	CANH	CAN High-Level Voltage I/O
8	8	8	8	STBY	Standby Mode Input
9	—	9	—	EP	Exposed Thermal Pad

1.7.1 TRANSMITTER DATA INPUT PIN (TxD)

The CAN transceiver drives the differential output pins, CANH and CANL, according to TxD. It is usually connected to the transmitter data output of the CAN controller device. When TxD is low, CANH and CANL are in the Dominant state. When TxD is high, CANH and CANL are in the Recessive state, provided that another CAN node is not driving the CAN bus with a Dominant state. TxD is connected from an internal pull-up resistor (nominal 33 kΩ) to VIO in the MCP2542FD and MCP2542WFD, and to VDD in the MCP2544FD and MCP2544WFDT.

1.7.2 GROUND SUPPLY PIN (VSS)

Ground supply pin.

1.7.3 SUPPLY VOLTAGE PIN (VDD)

Positive supply voltage pin. Supplies transmitter and receiver, including the wake-up receiver.

1.7.4 RECEIVER DATA OUTPUT PIN (RxD)

RxD is a CMOS-compatible output that drives high or low depending on the differential signals on the CANH and CANL pins, and is usually connected to the receiver data input of the CAN controller device. RxD is high when the CAN bus is Recessive and low in the Dominant state. RxD is supplied by VIO in the MCP2542FD and MCP2542WFD, and by VDD in the MCP2544FD and MCP2544WFDT.

1.7.5 NC PIN (MCP2544FD AND MCP2544WFDT)

No Connect. This pin can be left open or connected to VSS.

1.7.6 VIO PIN (MCP2542FD AND MCP2542WFD)

Supply for digital I/O pins. In the MCP2544FD and MCP2544WFDT, the supply for the digital I/O (TxD, RxD and STBY) is internally connected to VDD.

1.7.7 DIGITAL I/O

The MCP2542FD/4FD and MCP2542WFD/4WFD enable easy interfacing to MCU with I/O ranges from 1.8V to 5V.

1.7.7.1 MCP2544FD and MCP2544WFDT

The VIH(MIN) and VIL(MAX) for STBY and TxD are independent of VDD. They are set at levels that are compatible with 3V and 5V microcontrollers.

The RxD pin is always driven to VDD, therefore, a 3V microcontroller will need a 5V tolerant input.

1.7.7.2 MCP2542FD and MCP2542WFD

VIH and VIL for STBY and TxD depend on VIO. The RxD pin is driven to VIO.

MCP2542FD/4FD, MCP2542WFD/4WFD

1.7.8 CAN LOW PIN (CANL)

The CANL output drives the low side of the CAN differential bus. This pin is also tied internally to the receive input comparator. CANL disconnects from the bus when MCP2542FD/4FD and MCP2542WFD/4WFD are not powered.

1.7.9 CAN HIGH PIN (CANH)

The CANH output drives the high side of the CAN differential bus. This pin is also tied internally to the receive input comparator. CANH disconnects from the bus when MCP2542FD/4FD and MCP2542WFD/4WFD are not powered.

1.7.10 STANDBY MODE INPUT PIN (STBY)

This pin selects between Normal or Standby mode. In Standby mode, the transmitter and high-speed receiver are turned off, only the low-power receiver and wake-up filter are active. STBY is connected from an internal MOS pull-up resistor to V_{IO} in the MCP2542FD and MCP2542WFD, and to V_{DD} in the MCP2544FD and MCP2544WFD. The value of the MOS pull-up resistor depends on the supply voltage. Typical values are 660 k Ω for 5V, 1.1 M Ω for 3.3V and 4.4 M Ω for 1.8V.

1.7.11 EXPOSED THERMAL PAD (EP)

It is recommended to connect this pad to V_{SS} to enhance electromagnetic immunity and thermal resistance.

MCP2542FD/4FD, MCP2542WFD/4WFD

1.8 Typical Applications

In order to meet the EMC/EMI requirements, a Common-Mode Choke (CMC) may be required for data rates greater than 1 Mbps. [Figure 1-3](#) and [Figure 1-4](#) illustrate examples of typical applications of the devices.

FIGURE 1-3: MCP2544WFD with NC and SPLIT TERMINATION

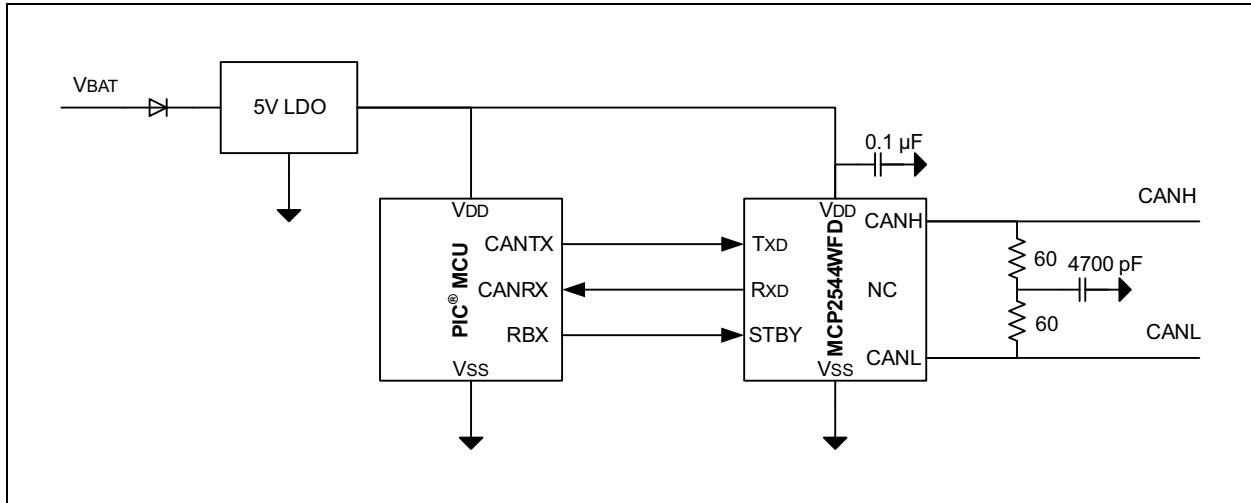
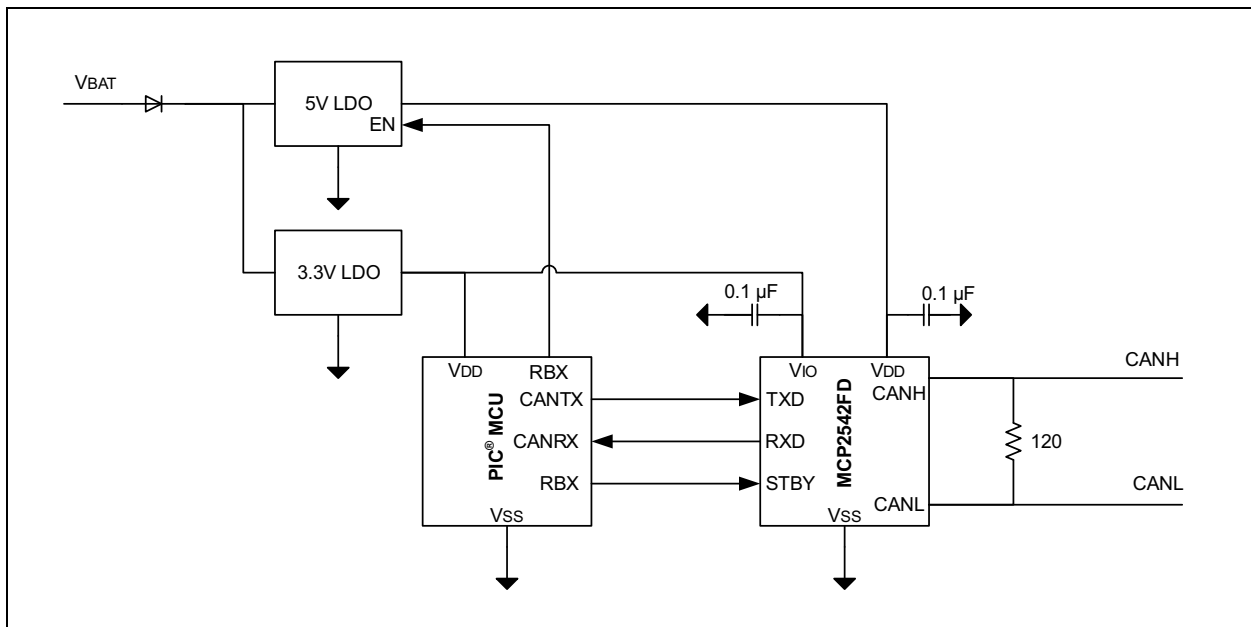


FIGURE 1-4: MCP2542FD with VIO PIN



MCP2542FD/4FD, MCP2542WFD/4WFD

NOTES:

MCP2542FD/4FD, MCP2542WFD/4WFD

2.0 ELECTRICAL CHARACTERISTICS

2.1 Terms and Definitions

A number of terms are defined in ISO 11898 that are used to describe the electrical characteristics of a CAN transceiver device. These terms and definitions are summarized in this section.

2.1.1 BUS VOLTAGE

V_{CANL} and V_{CANH} denote the voltages of the bus line wires, CANL and CANH, relative to the ground of each individual CAN node.

2.1.2 COMMON-MODE BUS VOLTAGE RANGE

Boundary voltage levels of V_{CANL} and V_{CANH} , with respect to ground, for which proper operation will occur if up to the maximum number of CAN nodes are connected to the bus.

2.1.3 DIFFERENTIAL INTERNAL CAPACITANCE, C_{DIFF} (OF A CAN NODE)

Capacitance seen between CANL and CANH during the Recessive state when the CAN node is disconnected from the bus (see [Figure 2-1](#)).

2.1.4 DIFFERENTIAL INTERNAL RESISTANCE, R_{DIFF} (OF A CAN NODE)

Resistance seen between CANL and CANH during the Recessive state when the CAN node is disconnected from the bus (see [Figure 2-1](#)).

2.1.5 DIFFERENTIAL VOLTAGE, V_{DIFF} (OF CAN BUS)

Differential voltage of the two-wire CAN bus with value equal to $V_{DIFF} = V_{CANH} - V_{CANL}$.

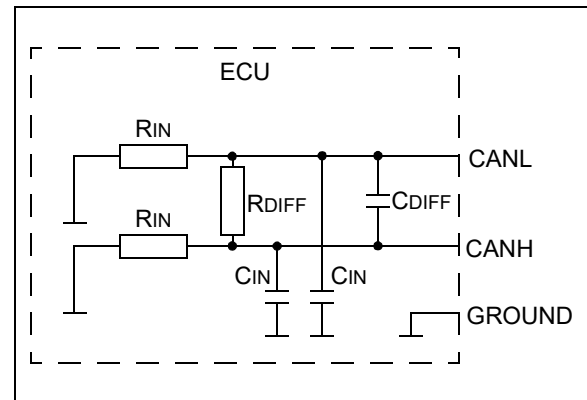
2.1.6 INTERNAL CAPACITANCE, C_{IN} (OF A CAN NODE)

Capacitance seen between CANL (or CANH) and ground during the Recessive state when the CAN node is disconnected from the bus (see [Figure 2-1](#)).

2.1.7 INTERNAL RESISTANCE, R_{IN} (OF A CAN NODE)

Resistance seen between CANL (or CANH) and ground during the Recessive state when the CAN node is disconnected from the bus (see [Figure 2-1](#)).

FIGURE 2-1: PHYSICAL LAYER DEFINITIONS



MCP2542FD/4FD, MCP2542WFD/4WFD

2.2 Absolute Maximum Ratings†

VDD	7.0V
VIO	7.0V
DC Voltage at TXD, RXD, STBY and VSS	-0.3V to VIO + 0.3V
DC Voltage at CANH and CANL	-58V to +58V
Transient Voltage on CANH and CANL (ISO-7637) (Figure 2-5)	-150V to +100V
Differential Bus Input Voltage VDIFF(I) (t = 60 days, continuous)	-5V to +10V
Differential Bus Input Voltage VDIFF(I) (1000 pulses, t = 0.1 ms, VCANH = +18V)	+17V
Dominant State Detection VDIFF(I) (10000 pulses, t = 1 ms)	+9V
Storage Temperature	-55°C to +150°C
Operating Ambient Temperature	-40°C to +150°C
Virtual Junction Temperature, TVJ (IEC 60747-1)	-40°C to +190°C
Soldering Temperature of Leads (10 seconds)	+300°C
ESD Protection on CANH and CANL Pins (IEC 61000-4-2)	±13 kV
ESD Protection on CANH and CANL Pins (IEC 801; Human Body Model)	±8 kV
ESD Protection on All Other Pins (IEC 801; Human Body Model)	±4 kV
ESD Protection on All Pins (IEC 801; Machine Model)	±400V
ESD Protection on All Pins (IEC 801; Charge Device Model)	±750V

† **Notice:** Stresses above those listed under “Maximum ratings” may cause permanent damage to the device. This is a stress rating only and functional operation of the device at those or any other conditions above those indicated in the operational listings of this specification is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability.

MCP2542FD/4FD, MCP2542WFD/4WFD

TABLE 2-1: DC CHARACTERISTICS

DC Specifications		Electrical Characteristics: Unless otherwise indicated, Extended (E): TAMB = -40°C to +125°C and High (H): TAMB = -40°C to +150°C; VDD = 4.5V to 5.5V; VIO = 1.7V to 5.5V (Note 2); RL = 60Ω; CL = 100 pF; unless otherwise specified.				
Parameter	Sym.	Min.	Typ.	Max.	Units	Conditions
Supply						
VDD Pin						
Voltage Range	VDD	4.5	—	5.5	V	
Supply Current	IDD	—	2.5	5	mA	Recessive, VTXD = VDD
		—	55	70		Dominant, VTXD = 0V
Standby Current	IDDS	—	4	15	μA	MCP2544FD and MCP2544WFD , bus recessive
		—	4	16		MCP2542FD and MCP2542WFD , includes IIO
Maximum Supply Current	IDDMAX	—	95	140	mA	Fault condition: VTXD = VSS, VCANH = VCANL = -5V to +18V (Note 1)
High Level of the POR Comparator for VDD	VPORH	—	3.0	3.95	V	Note 1
Low Level of the POR Comparator for VDD	VPORL	1.0	2.0	3.2	V	Note 1
Hysteresis of POR Comparator for VDD	VPORD	0.2	0.9	2.0	V	Note 1
High Level of the UV Comparator for VDD	VUVH	4.0	4.25	4.4	V	
Low Level of the UV Comparator for VDD	VUVL	3.6	3.8	4.0	V	
Hysteresis of UV Comparator	VUVD	—	0.4	—	V	Note 1
VIO Pin						
Digital Supply Voltage Range	VIO	1.7	—	5.5	V	
Supply Current on VIO	IIO	—	7	20	μA	Recessive, VTXD = VIO
		—	200	400		Dominant, VTXD = 0V
Standby Current	IDDS	—	0.3	2	μA	Bus recessive (Note 1)
High Level of the POR Comparator for VIO	VPORH_VIO	0.8	1.2	1.7	V	
Low Level of the POR Comparator for VIO	VPORL_VIO	0.7	1.1	1.4	V	
Hysteresis of POR Comparator for VIO	VPORD_VIO	—	0.2	—	V	
Bus Line (CANH; CANL) Transmitter						
CANH; CANL: Recessive Bus Output Voltage	VO(R)	2.0	0.5 VDD	3.0	V	VTXD = VDD, no load

Note 1: Characterized; not 100% tested.

2: Only MCP2542FD and MCP2542WFD have a VIO pin. For the MCP2544FD and MCP2544WFD, VIO is internally connected to VDD.

3: -12V to 12V is ensured by characterization and tested from -2V to 7V.

MCP2542FD/4FD, MCP2542WFD/4WFD

TABLE 2-1: DC CHARACTERISTICS (CONTINUED)

DC Specifications	Electrical Characteristics: Unless otherwise indicated, Extended (E): T _{AMB} = -40°C to +125°C and High (H): T _{AMB} = -40°C to +150°C; V _{DD} = 4.5V to 5.5V; V _{IO} = 1.7V to 5.5V (Note 2); R _L = 60Ω; C _L = 100 pF; unless otherwise specified.					
Parameter	Sym.	Min.	Typ.	Max.	Units	Conditions
CANH; CANL: Bus Output Voltage in Standby	V _{O(S)}	-0.1	0.0	+0.1	V	STBY = V _{TXD} = V _{DD} , no load
Recessive Output Current	I _{O(R)}	-5	—	+5	mA	-24V < V _{CAN} < +24V
CANH: Dominant Output Voltage	V _{O(D)}	2.75	3.50	4.50	V	T _{XD} = 0, R _L = 50 to 65Ω
CANL: Dominant Output Voltage		0.50	1.50	2.25		R _L = 50Ω to 65Ω
Driver Symmetry (V _{CANH} + V _{CANL})/V _{DD}	V _{SYM}	0.9	1.0	1.1	V	1 MHz square wave, Recessive and Dominant states, and transition (Note 1)
Dominant: Differential Output Voltage	V _{O(DIFF)} (D)	1.5	2.0	3.0	V	V _{TXD} = V _{SS} , R _L = 50Ω to 65Ω (Figure 2-2, Figure 2-4, Section 3.0 “Typical Performance Curves”) (Note 1)
		1.4	2.0	3.3		V _{TXD} = V _{SS} , R _L = 45Ω to 70Ω (Figure 2-2, Figure 2-4, Section 3.0 “Typical Performance Curves”) (Note 1)
		1.3	2.0	3.3		V _{TXD} = V _{SS} , R _L = 40Ω to 75Ω (Figure 2-2, Figure 2-4)
		1.5	—	5.0		V _{TXD} = V _{SS} , R _L = 2240Ω (Figure 2-2, Figure 2-4, Section 3.0 “Typical Performance Curves”) (Note 1)
Recessive: Differential Output Voltage	V _{O(DIFF)} (R)	-500	0	50	mV	V _{TXD} = V _{DD} , no load, normal (Figure 2-2, Figure 2-4)
	V _{O(DIFF)} (S)	-200	0	200		V _{TXD} = V _{DD} , no load, standby Figure 2-2, Figure 2-4
CANH: Short-Circuit Output Current	I _{O(SC)}	-115	-85	—	mA	V _{TXD} = V _{SS} , V _{CANH} = -3V, CANL: floating
CANL: Short-Circuit Output Current		—	75	+115	mA	V _{TXD} = V _{SS} , V _{CANL} = +18V, CANH: floating
Bus Line (CANH; CANL) Receiver						
Recessive Differential Input Voltage	V _{DIFF(R)} (I)	-4.0	—	+0.5	V	Normal mode, -12V < V(CANH, CANL) < +12V, see Figure 2-6 (Note 3)
		-4.0	—	+0.4		Standby mode, -12V < V(CANH, CANL) < +12V, see Figure 2-6 (Note 3)

Note 1: Characterized; not 100% tested.

2: Only MCP2542FD and MCP2542WFD have a V_{IO} pin. For the MCP2544FD and MCP2544WFDT, V_{IO} is internally connected to V_{DD}.

3: -12V to 12V is ensured by characterization and tested from -2V to 7V.

MCP2542FD/4FD, MCP2542WFD/4WFD

TABLE 2-1: DC CHARACTERISTICS (CONTINUED)

DC Specifications		Electrical Characteristics: Unless otherwise indicated, Extended (E): TAMB = -40°C to +125°C and High (H): TAMB = -40°C to +150°C; VDD = 4.5V to 5.5V; VIO = 1.7V to 5.5V (Note 2); RL = 60Ω; CL = 100 pF; unless otherwise specified.				
Parameter	Sym.	Min.	Typ.	Max.	Units	Conditions
Dominant Differential Input Voltage	VDIFF(D)(I)	0.9	—	9.0	V	Normal mode, -12V < V(CANH, CANL) < +12V, see Figure 2-6 (Note 3)
		1.1	—	9.0		Standby mode, -12V < V(CANH, CANL) < +12V, see Figure 2-6 (Note 3)
Differential Receiver Threshold	VTH(DIFF)	0.5	0.7	0.9	V	Normal mode, -12V < V(CANH, CANL) < +12V, see Figure 2-6 (Note 3)
		0.4	0.7	0.9		Standby mode, -12V < V(CANH, CANL) < +12V, see Figure 2-6 (Note 3)
Differential Input Hysteresis	VHYS(DIFF)	30	—	200	mV	Normal mode, see Figure 2-6 (Note 1)
Single-Ended Input Resistance	RCAN_H, RCAN_L	6	—	50	kΩ	Note 1
Internal Resistance Matching $mR = 2 * (RCANH - RCANL) / (RCANH + RCANL)$	mR	-3	0	+3	%	VCANH = VCANL (Note 1)
Differential Input Resistance	RDIFF	12	25	100	kΩ	Note 1
Internal Capacitance	CIN	—	20	—	pF	1 Mbps (Note 1)
Differential Internal Capacitance	CDIFF	—	10	—	pF	1 Mbps (Note 1)
CANH, CANL: Input Leakage	ILI	-5	—	+5	μA	VDD = VTXD = VSTBY = 0V, for MCP2542FD and MCP2542WFD , VIO = 0V, VCANH = VCANL = 5 V
Digital Input Pins (TXD, STBY)						
High-Level Input Voltage	VIH	2.0	—	VDD + 0.3	V	MCP2544FD and MCP2544WFDT
		0.7 VIO	—	VIO + 0.3		MCP2542FD and MCP2542WFD
Low-Level Input Voltage	VIL	-0.3	—	0.8	V	MCP2544FD and MCP2544WFDT
		-0.3	—	0.3 VIO		MCP2542FD and MCP2542WFD
High-Level Input Current	IIH	-1	—	+1	μA	
TXD: Low-Level Input Current	IIL(TXD)	-270	-150	-30	μA	
STBY: Low-Level Input Current	IIL(STBY)	-30	—	-1	μA	

Note 1: Characterized; not 100% tested.

2: Only MCP2542FD and MCP2542WFD have a VIO pin. For the MCP2544FD and MCP2544WFDT, VIO is internally connected to VDD.

3: -12V to 12V is ensured by characterization and tested from -2V to 7V.

MCP2542FD/4FD, MCP2542WFD/4WFD

TABLE 2-1: DC CHARACTERISTICS (CONTINUED)

DC Specifications	Electrical Characteristics: Unless otherwise indicated, Extended (E): T _{AMB} = -40°C to +125°C and High (H): T _{AMB} = -40°C to +150°C; V _{DD} = 4.5V to 5.5V; V _{IO} = 1.7V to 5.5V (Note 2); R _L = 60Ω; C _L = 100 pF; unless otherwise specified.					
Parameter	Sym.	Min.	Typ.	Max.	Units	Conditions
Receive Data (RxD) Output						
High-Level Output Voltage	VOH	V _{DD} - 0.4	—	—	V	MCP2544FD and MCP2544WFDT: I _{OH} = -2 mA, typical -4 mA
		V _{IO} - 0.4	—	—		MCP2542FD and MCP2542WFD: V _{IO} = 2.7V to 5.5V, I _{OH} = -1 mA, V _{IO} = 1.7V to 2.7V, I _{OH} = -0.5 mA, typical -2 mA
Low-Level Output Voltage	VOL	—	—	0.4	V	I _{OL} = 4 mA, typical 8 mA
Thermal Shutdown						
Shutdown Junction Temperature	T _{J(SD)}	165	175	185	°C	-12V < V(CANH, CANL) < +12V (Note 1)
Shutdown Temperature Hysteresis	T _{J(HYST)}	15	—	30	°C	-12V < V(CANH, CANL) < +12V (Note 1)

Note 1: Characterized; not 100% tested.

2: Only MCP2542FD and MCP2542WFD have a V_{IO} pin. For the MCP2544FD and MCP2544WFDT, V_{IO} is internally connected to V_{DD}.

3: -12V to 12V is ensured by characterization and tested from -2V to 7V.

MCP2542FD/4FD, MCP2542WFD/4WFD

FIGURE 2-2: PHYSICAL BIT REPRESENTATION AND SIMPLIFIED BIAS IMPLEMENTATION

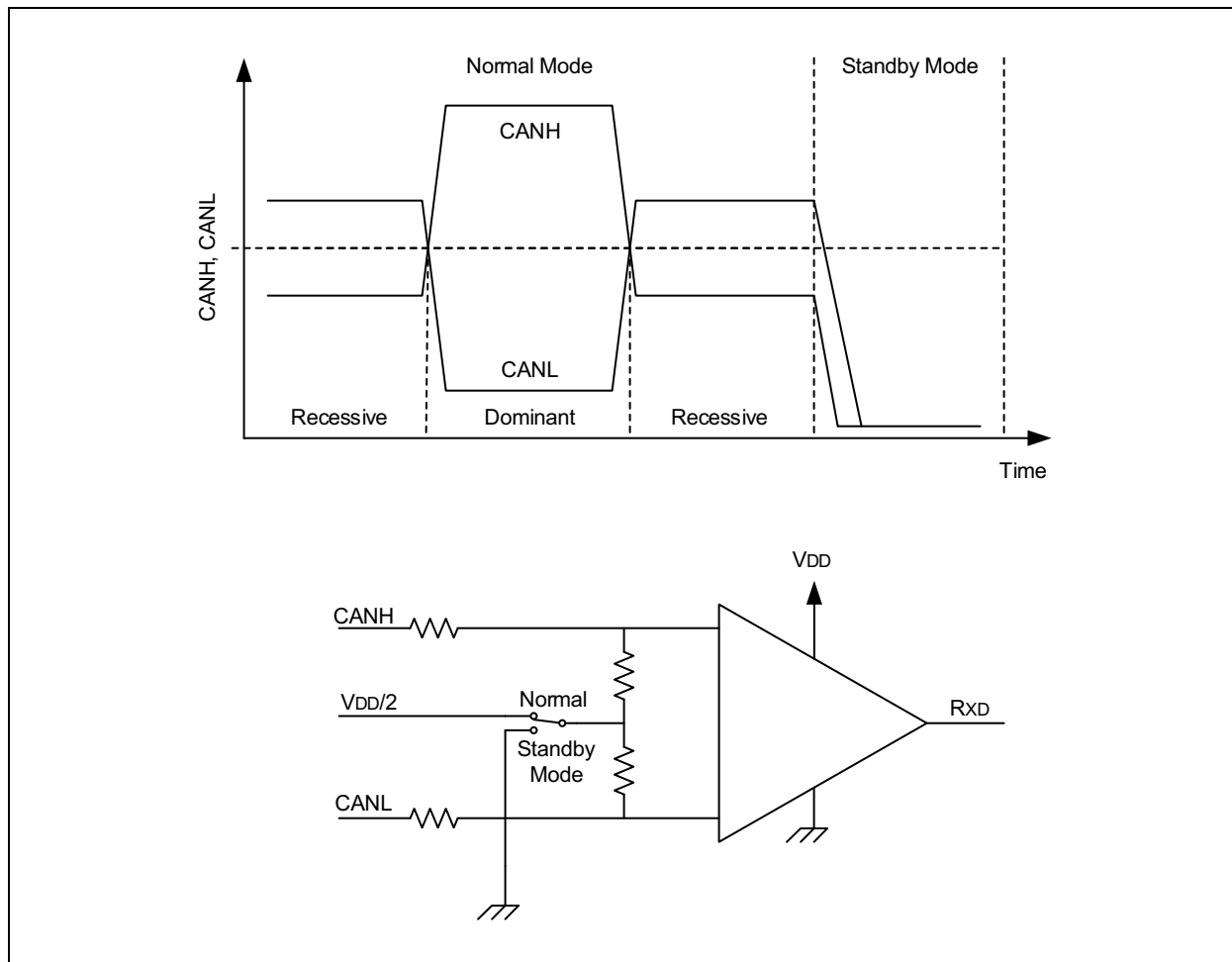


TABLE 2-2: AC CHARACTERISTICS

AC Characteristics		Electrical Characteristics: Unless otherwise indicated, Extended (E): T _{AMB} = -40°C to +125°C and High (H): T _{AMB} = -40°C to +150°C; V _{DD} = 4.5V to 5.5V; V _{IO} = 1.7V to 5.5V (Note 2); R _L = 60Ω; C _L = 100 pF; Maximum V _{DIFF(D)} (I) = 3V.					
Param. No.	Parameter	Sym.	Min.	Typ.	Max.	Units	Conditions
1	Bit Time	t _{BIT}	0.125	—	69.44	μs	
2	Nominal Bit Rate	NBR	14.4	—	8000	kbps	
3	Delay TXD Low to Bus Dominant	t _{TXD-BUSON}	—	50	85	ns	Note 1
4	Delay TXD High to Bus Recessive	t _{TXD-BUSOFF}	—	40	85	ns	Note 1
5	Delay Bus Dominant to RXD	t _{BUSON-RXD}	—	70	85	ns	Note 1
6	Delay Bus Recessive to RXD	t _{BUSOFF-RXD}	—	110	145	ns	Note 1

Note 1: Characterized, not 100% tested.

2: Only MCP2542FD and MCP2542WFD have a V_{IO} pin. For the MCP2544FD and MCP2544WFD, V_{IO} is internally connected to V_{DD}.

3: Characterized. Not in ISO 11898-2:2016.

MCP2542FD/4FD, MCP2542WFD/4WFD

TABLE 2-2: AC CHARACTERISTICS (CONTINUED)

AC Characteristics		Electrical Characteristics: Unless otherwise indicated, Extended (E): TAMB = -40°C to +125°C and High (H): TAMB = -40°C to +150°C; VDD = 4.5V to 5.5V; VIO = 1.7V to 5.5V (Note 2); RL = 60Ω; CL = 100 pF; Maximum VDIFF(D)(I) = 3V.					
Param. No.	Parameter	Sym.	Min.	Typ.	Max.	Units	Conditions
7	Propagation Delay TXD to RXD Worst Case of tLOOP(R) and tLOOP(F); see Figure 2-10	tTXD – RXD	—	90	120	ns	RL = 150Ω, CL = 200 pF (Note 1)
			—	115	150		
7a	Propagation Delay, Rising Edge	tLOOP(R)	—	90	120	ns	
7b	Propagation Delay, Falling Edge	tLOOP(F)	—	80	120	ns	
8a	Recessive Bit Time on RXD – 1 Mbps, Loop Delay Symmetry (Note 3)	tBIT(RXD), 1M	900	985	1100	ns	tBIT(TXD) = 1000 ns (Figure 2-10)
			800	960	1255		tBIT(TXD) = 1000 ns (Figure 2-10), RL = 150Ω, CL = 200 pF (Note 1)
8b	Recessive Bit Time on RXD – 2 Mbps, Loop Delay Symmetry	tBIT(RXD), 2M	450	490	550	ns	tBIT(TXD) = 500 ns (Figure 2-10)
			400	460	550		tBIT(TXD) = 500 ns (Figure 2-10), RL = 150Ω, CL = 200 pF (Note 1)
8c	Recessive Bit Time on RXD – 5 Mbps, Loop Delay Symmetry	tBIT(RXD), 5M	160	190	220	ns	tBIT(TXD) = 200 ns (Figure 2-10)
8d	Recessive Bit Time on RXD – 8 Mbps, Loop Delay Symmetry (Note 3)	tBIT(RXD), 8M	85	100	135	ns	tBIT(TXD) = 120 ns (Figure 2-10) (Note 1)
9	CAN Activity Filter Time (Standby)	tFILTER	0.5	1.7	3.6	μs	VDIFF(D)(I) = 1.2V to 3V
10	Delay Standby to Normal Mode	twAKE	—	7	30	μs	Negative edge on STBY
11	Permanent Dominant Detect Time	tPDT	0.8	1.9	5	ms	TXD = 0V
12	Permanent Dominant Timer Reset	tPDTR	—	5	—	ns	The shortest recessive pulse on TXD or CAN bus to reset permanent dominant timer

Note 1: Characterized, not 100% tested.

Note 2: Only MCP2542FD and MCP2542WFD have a VIO pin. For the MCP2544FD and MCP2544WFD, VIO is internally connected to VDD.

Note 3: Characterized. Not in ISO 11898-2:2016.

MCP2542FD/4FD, MCP2542WFD/4WFD

TABLE 2-2: AC CHARACTERISTICS (CONTINUED)

AC Characteristics		Electrical Characteristics: Unless otherwise indicated, Extended (E): TAMB = -40°C to +125°C and High (H): TAMB = -40°C to +150°C; VDD = 4.5V to 5.5V; VIO = 1.7V to 5.5V (Note 2); RL = 60Ω; CL = 100 pF; Maximum VDIFF(D)(I) = 3V.					
Param. No.	Parameter	Sym.	Min.	Typ.	Max.	Units	Conditions
13a	Transmitted Bit Time on Bus – 1 Mbps (Note 3)	tBIT(BUS), 1M	870	1000	1060	ns	tBIT(TXD) = 1000 ns (Figure 2-10)
			870	1000	1060		tBIT(TXD) = 1000 ns (Figure 2-10), RL = 150Ω, CL = 200 pF (Note 1)
13b	Transmitted Bit Time on Bus – 2 Mbps	tBIT(BUS), 2M	435	515	530	ns	tBIT(TXD) = 500 ns (Figure 2-10)
			435	480	550		tBIT(TXD) = 500 ns (Figure 2-10), RL = 150Ω, CL = 200 pF (Note 1)
13c	Transmitted Bit Time on Bus – 5 Mbps	tBIT(BUS), 5M	155	200	210	ns	tBIT(TXD) = 200ns (Figure 2-10) (Note 1)
13d	Transmitted Bit Time on Bus – 8 Mbps (Note 3)	tBIT(BUS), 8M	100	125	140	ns	tBIT(TXD) = 120 ns (Figure 2-10) (Note 1)
14a	Receiver Timing Symmetry – 1 Mbps (Note 3)	tDIFF(REC), 1M = tBIT(RXD) – tBIT(BUS)	-65	0	40	ns	tBIT(TXD) = 1000 ns (Figure 2-10)
			-130	0	80		tBIT(TXD) = 1000ns (Figure 2-10), RL = 150Ω, CL = 200 pF (Note 1)
14b	Receiver Timing Symmetry – 2 Mbps	tDIFF(REC), 2M	-65	0	40	ns	tBIT(TXD) = 500 ns (Figure 2-10)
			-70	0	40		tBIT(TXD) = 500 ns (Figure 2-10), RL = 150Ω, CL = 200 pF (Note 1)
14c	Receiver Timing Symmetry – 5 Mbps	tDIFF(REC), 5M	-45	0	15	ns	tBIT(TXD) = 200 ns (Figure 2-10) (Note 1)
14d	Receiver Timing Symmetry – 8 Mbps (Note 3), tDIFF(REC), 8M	tDIFF(REC), 8M	-45	0	10	ns	tBIT(TXD) = 120 ns (Figure 2-10) (Note 1)
15	WUP Time-out	twAKE(TO)	1	1.9	5	ms	MCP2542WFD/4WFD (Figure 2-11)
16	Delay Bus Dominant/Recessive to RXD (Standby mode)	tBUS-RXD(S)	—	0.5	—	μs	

Note 1: Characterized, not 100% tested.

2: Only MCP2542FD and MCP2542WFD have a VIO pin. For the MCP2544FD and MCP2544WFD, VIO is internally connected to VDD.

3: Characterized. Not in ISO 11898-2:2016.

MCP2542FD/4FD, MCP2542WFD/4WFD

FIGURE 2-3: TEST LOAD CONDITIONS

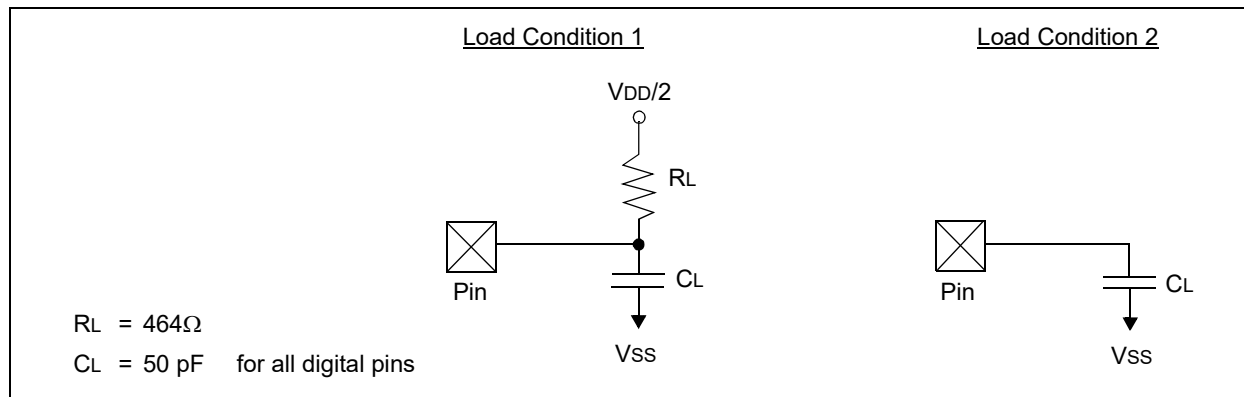


FIGURE 2-4: TEST CIRCUIT FOR ELECTRICAL CHARACTERISTICS

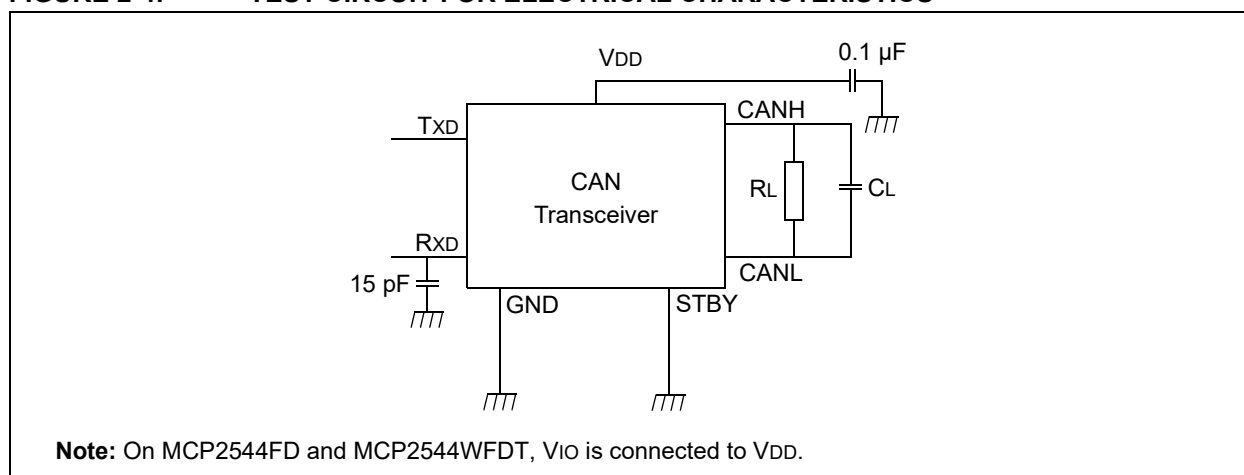
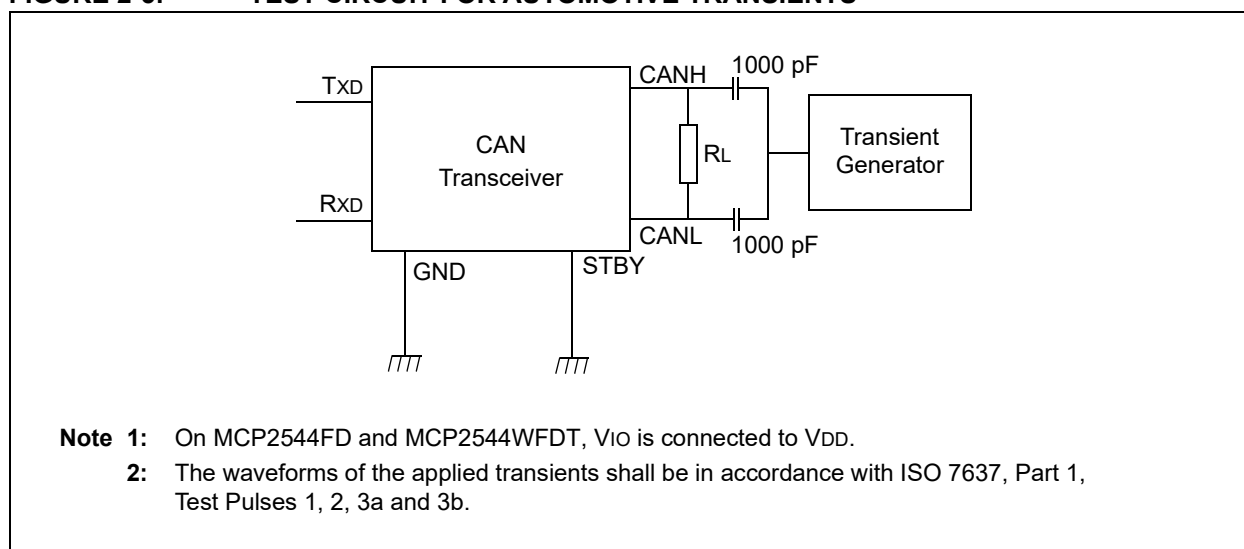
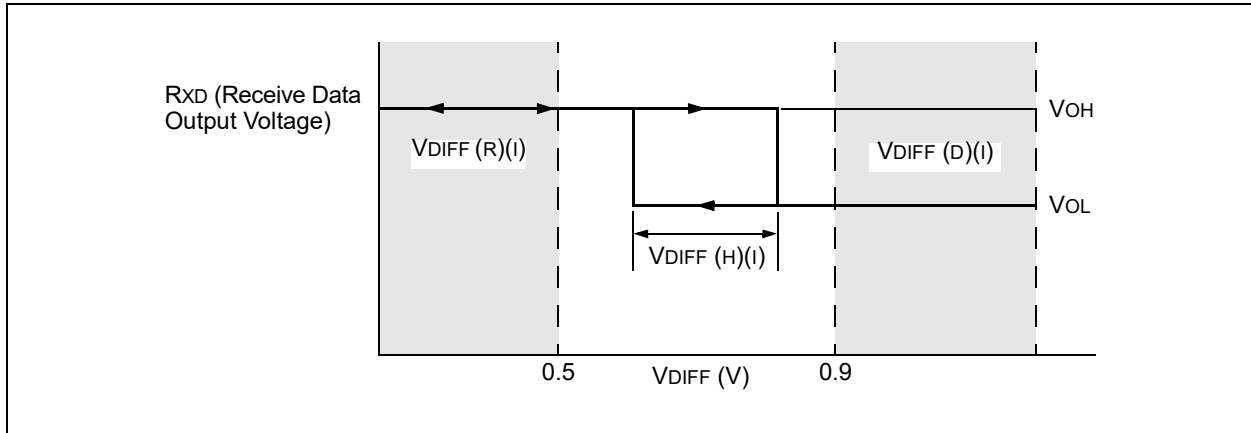


FIGURE 2-5: TEST CIRCUIT FOR AUTOMOTIVE TRANSIENTS^(1,2)



MCP2542FD/4FD, MCP2542WFD/4WFD

FIGURE 2-6: HYSTERESIS OF THE RECEIVER



MCP2542FD/4FD, MCP2542WFD/4WFD

2.3 Timing Diagrams and Specifications

FIGURE 2-7: TIMING DIAGRAM FOR AC CHARACTERISTICS

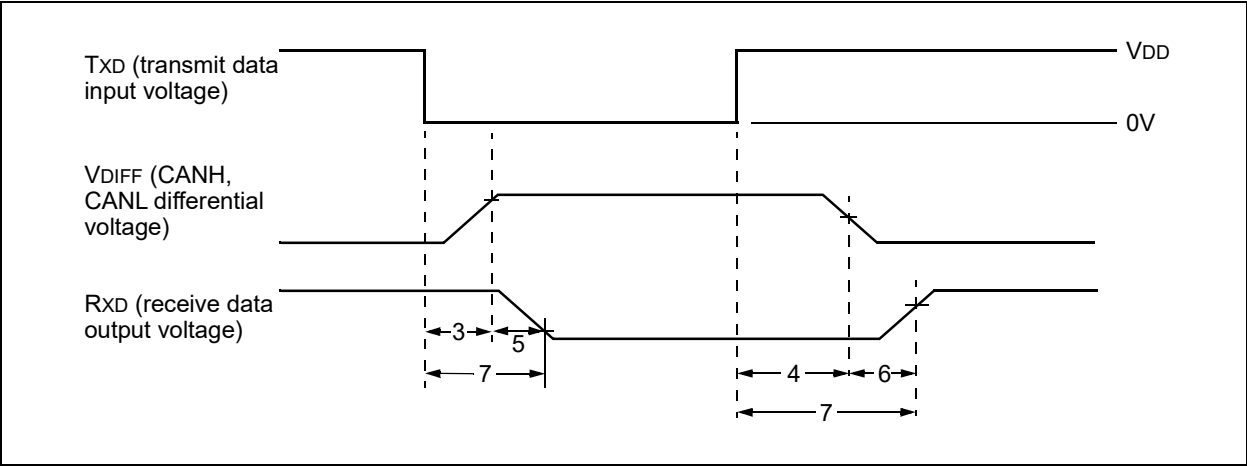


FIGURE 2-8: TIMING DIAGRAM FOR WAKE-UP FROM STANDBY

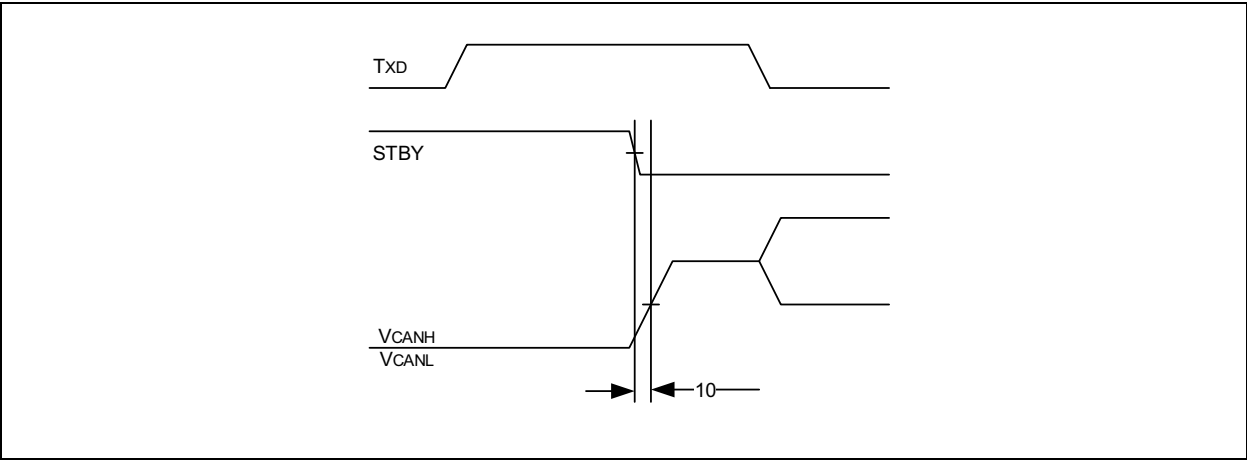
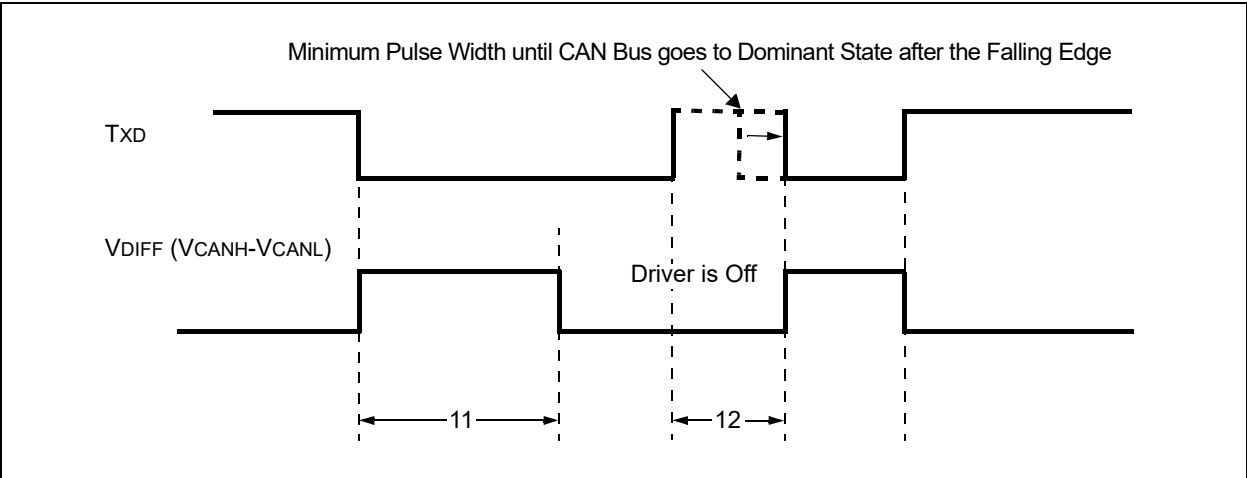


FIGURE 2-9: PERMANENT DOMINANT TIMER RESET DETECT



MCP2542FD/4FD, MCP2542WFD/4WFD

FIGURE 2-10: TIMING DIAGRAM FOR LOOP DELAY SYMMETRY

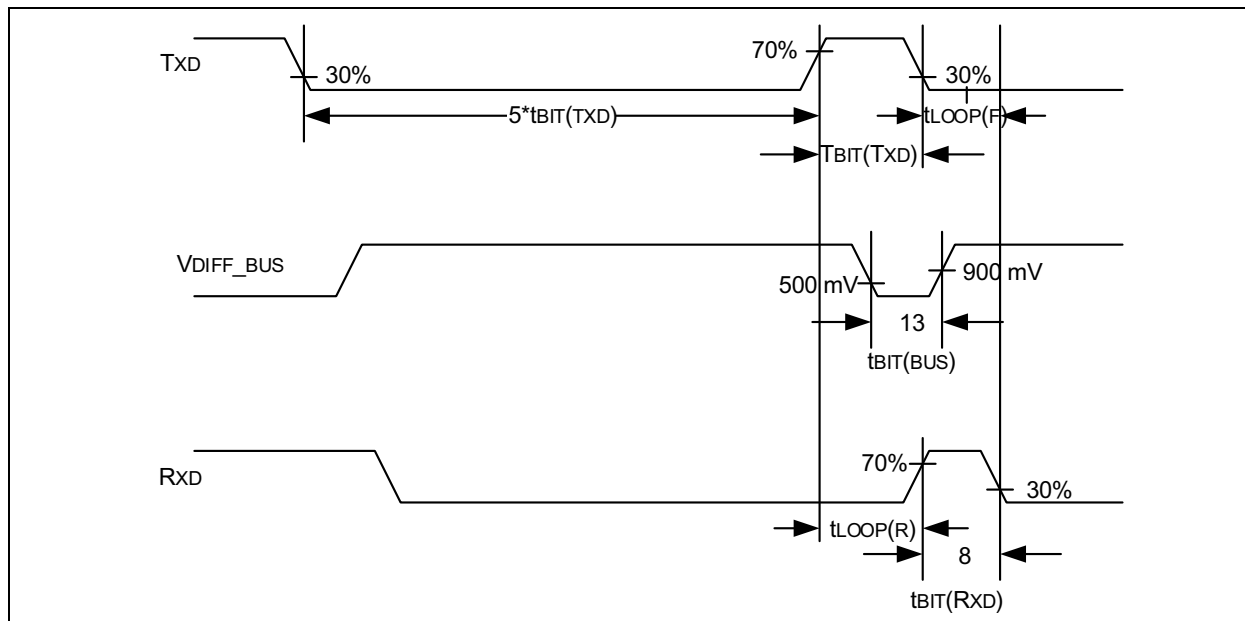


FIGURE 2-11: TIMING DIAGRAM FOR WAKE-UP PATTERN (WUP)

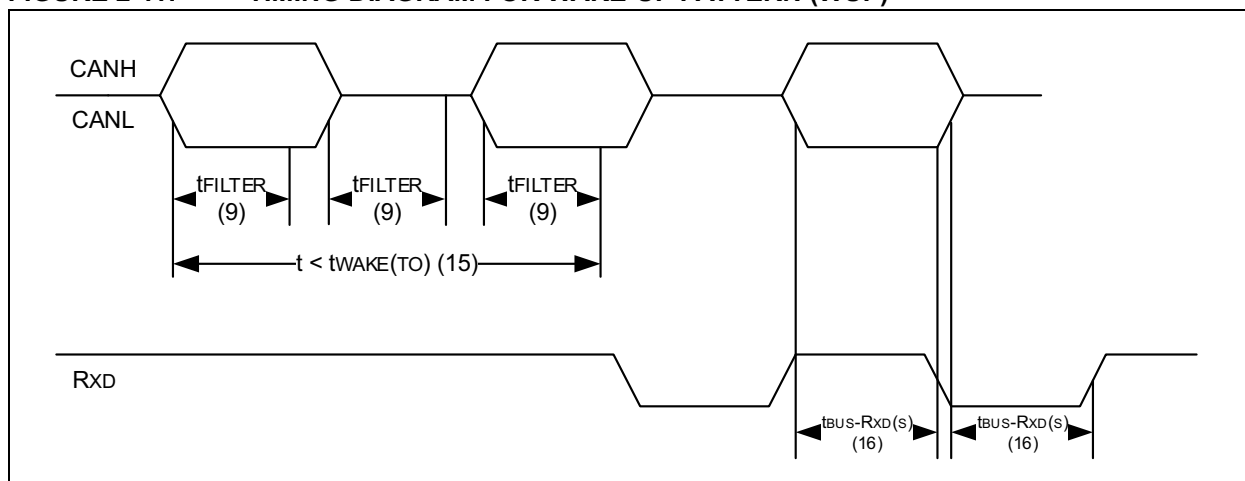


TABLE 2-3: THERMAL SPECIFICATIONS

Parameter	Sym.	Min.	Typ.	Max.	Units	Test Conditions
Temperature Ranges						
Specified Temperature Range	TA	-40	—	+125	°C	
		-40	—	+150		
Operating Temperature Range	TA	-40	—	+150	°C	
Storage Temperature Range	TA	-65	—	+155	°C	
Package Thermal Resistances						
Thermal Resistance, 8-Lead DFN (3x3)	θ_{JA}	—	56.7	—	°C/W	
Thermal Resistance, 8-Lead SOIC	θ_{JA}	—	149.5	—	°C/W	
Thermal Resistance, 8-Lead TDFN (2x3)	θ_{JA}	—	53	—	°C/W	

MCP2542FD/4FD, MCP2542WFD/4WFD

NOTES:

MCP2542FD/4FD, MCP2542WFD/4WFD

3.0 TYPICAL PERFORMANCE CURVES

Note: The graphs and tables provided following this note are a statistical summary based on a limited number of samples and are provided for informational purposes only. The performance characteristics listed herein are not tested or guaranteed. In some graphs or tables, the data presented may be outside the specified operating range (e.g., outside specified power supply range) and therefore outside the warranted range.

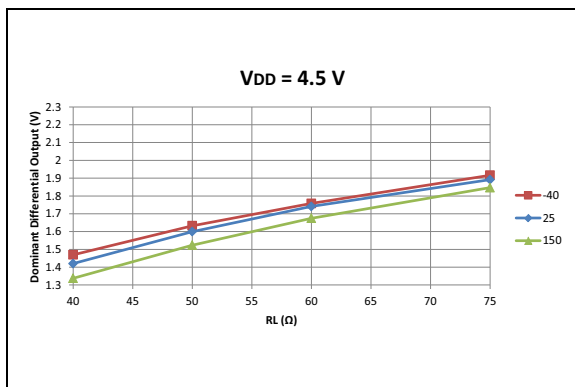


FIGURE 3-1: Dominant Differential Output vs. R_L ($V_{DD} = 4.5V$).

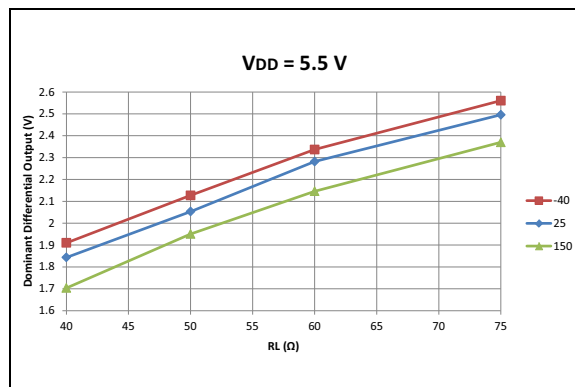


FIGURE 3-3: Dominant Differential Output vs. R_L ($V_{DD} = 5.5V$).

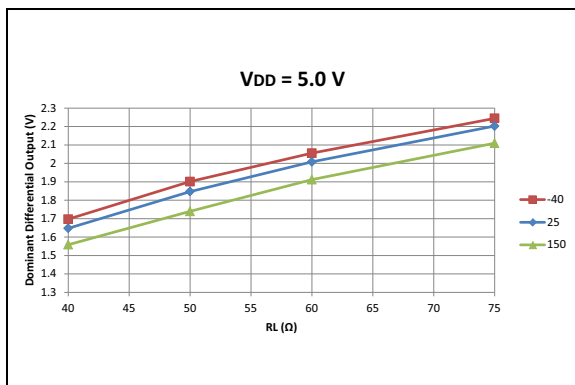


FIGURE 3-2: Dominant Differential Output vs. R_L ($V_{DD} = 5.0V$).

MCP2542FD/4FD, MCP2542WFD/4WFD

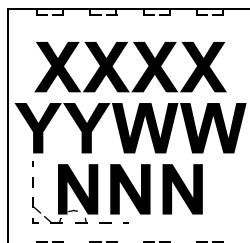
NOTES:

MCP2542FD/4FD, MCP2542WFD/4WFD

4.0 PACKAGING INFORMATION

4.1 Package Marking Information

8-Lead DFN (03x03x0.9 mm)

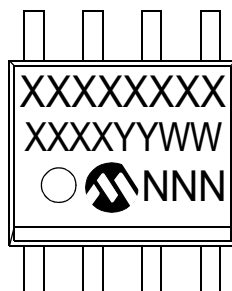


Part Number	Code
MCP2542FD-E/MF	DAEK
MCP2542FDT-H/MF	DAEK
MCP2542FD-H/MF	DAEK
MCP2542FDT-E/MF	DAEK
MCP2542WFD-E/MF	DAEH
MCP2542WFDT-H/MF	DAEH
MCP2542WFD-H/MF	DAEH
MCP2542WFDT-E/MF	DAEH
MCP2544FD-E/MF	DAEJ
MCP2544FDT-H/MF	DAEJ
MCP2544FD-H/MF	DAEJ
MCP2544FDT-E/MF	DAEJ
MCP2544WFD-E/MF	DAEG
MCP2544WFDT-H/MF	DAEG
MCP2544WFD-H/MF	DAEG
MCP2544WFDT-E/MF	DAEG

Example

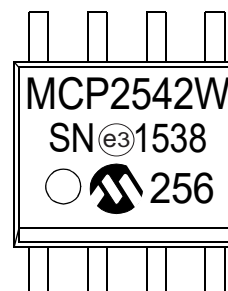


8-Lead SOIC (150 mil)



Part Number	Code
MCP2542WFD-E/SN	MCP2542W
MCP2542WFDT-H/SN	MCP2542W
MCP2542WFD-H/SN	MCP2542W
MCP2542WFDT-E/SN	MCP2542W
MCP2542FD-E/SN	MCP2542
MCP2542FDT-H/SN	MCP2542
MCP2542FD-H/SN	MCP2542
MCP2542FDT-E/SN	MCP2542
MCP2544WFD-E/SN	MCP2544W
MCP2544WFDT-H/SN	2544WFD
MCP2544WFD-H/SN	2544WFD
MCP2544WFDT-E/SN	MCP2544W
MCP2544FD-E/SN	MCP2544
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MCP2544FD-H/SN	MCP2544
MCP2544FDT-E/SN	MCP2544

Example

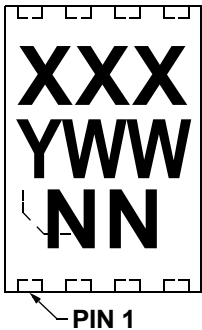


Legend:	XX...X	Customer-specific information
	Y	Year code (last digit of calendar year)
	YY	Year code (last 2 digits of calendar year)
	WW	Week code (week of January 1 is week '01')
	NNN	Alphanumeric traceability code
	(e3)	Pb-free JEDEC® designator for Matte Tin (Sn)
		This package is Pb-free. The Pb-free JEDEC® designator (e3) can be found on the outer packaging for this package.
Note:	In the event the full Microchip part number cannot be marked on one line, it will be carried over to the next line, thus limiting the number of available characters for customer-specific information.	

MCP2542FD/4FD, MCP2542WFD/4WFD

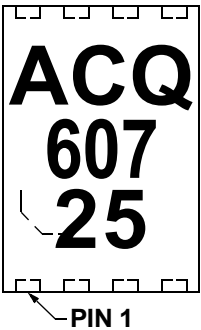
4.1 Package Marking Information (Continued)

8-Lead TDFN (02x03x0.8 mm)



Part Number	Code
MCP2542FDT-E/MNY	ACR
MCP2542FDT-H/MNY	ACR
MCP2542WFDT-E/MNY	ACP
MCP2542WFDT-H/MNY	ACP
MCP2544FDT-E/MNY	ACQ
MCP2544FDT-H/MNY	ACQ
MCP2544WFDTT-E/MNY	ACN
MCP2544WFDTT-H/MNY	ACN

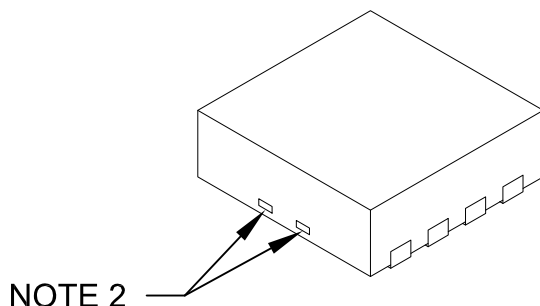
Example



MCP2542FD/4FD, MCP2542WFD/4WFD

8-Lead Plastic Dual Flat, No Lead Package (MF) - 3x3x0.9mm Body [DFN]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



		Units	MILLIMETERS		
Dimension Limits			MIN	NOM	MAX
Number of Pins	N		8		
Pitch	e		0.65 BSC		
Overall Height	A		0.80	0.90	1.00
Standoff	A1		0.00	0.02	0.05
Contact Thickness	A3		0.20 REF		
Overall Length	D		3.00 BSC		
Exposed Pad Width	E2		1.34	-	1.60
Overall Width	E		3.00 BSC		
Exposed Pad Length	D2		1.60	-	2.40
Contact Width	b		0.25	0.30	0.35
Contact Length	L		0.20	0.30	0.55
Contact-to-Exposed Pad	K		0.20	-	-

Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Package may have one or more exposed tie bars at ends.
3. Package is saw singulated
4. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

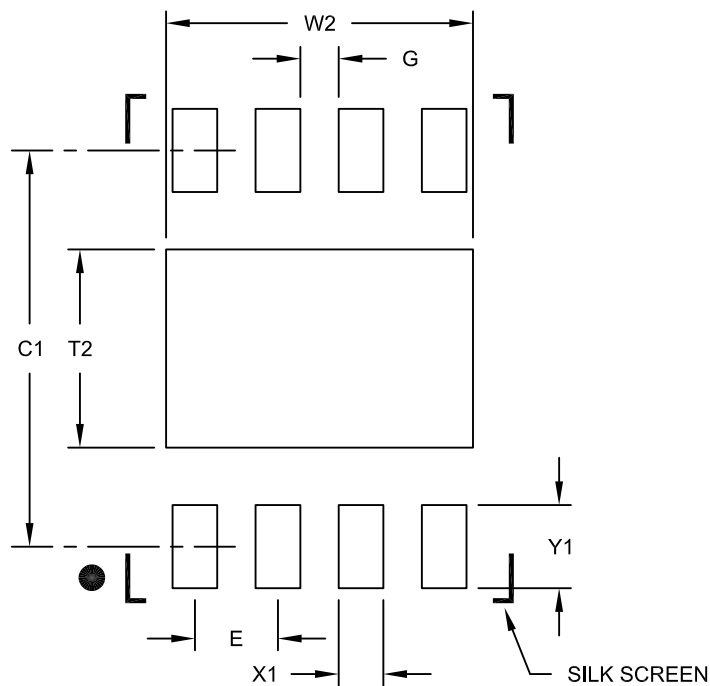
REF: Reference Dimension, usually without tolerance, for information purposes only.

Microchip Technology Drawing No. C04-062C Sheet 2 of 2

MCP2542FD/4FD, MCP2542WFD/4WFD

8-Lead Plastic Dual Flat, No Lead Package (MF) - 3x3x0.9mm Body [DFN]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.65 BSC		
Optional Center Pad Width	W2			2.40
Optional Center Pad Length	T2			1.55
Contact Pad Spacing	C1		3.10	
Contact Pad Width (X8)	X1			0.35
Contact Pad Length (X8)	Y1			0.65
Distance Between Pads	G	0.30		

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

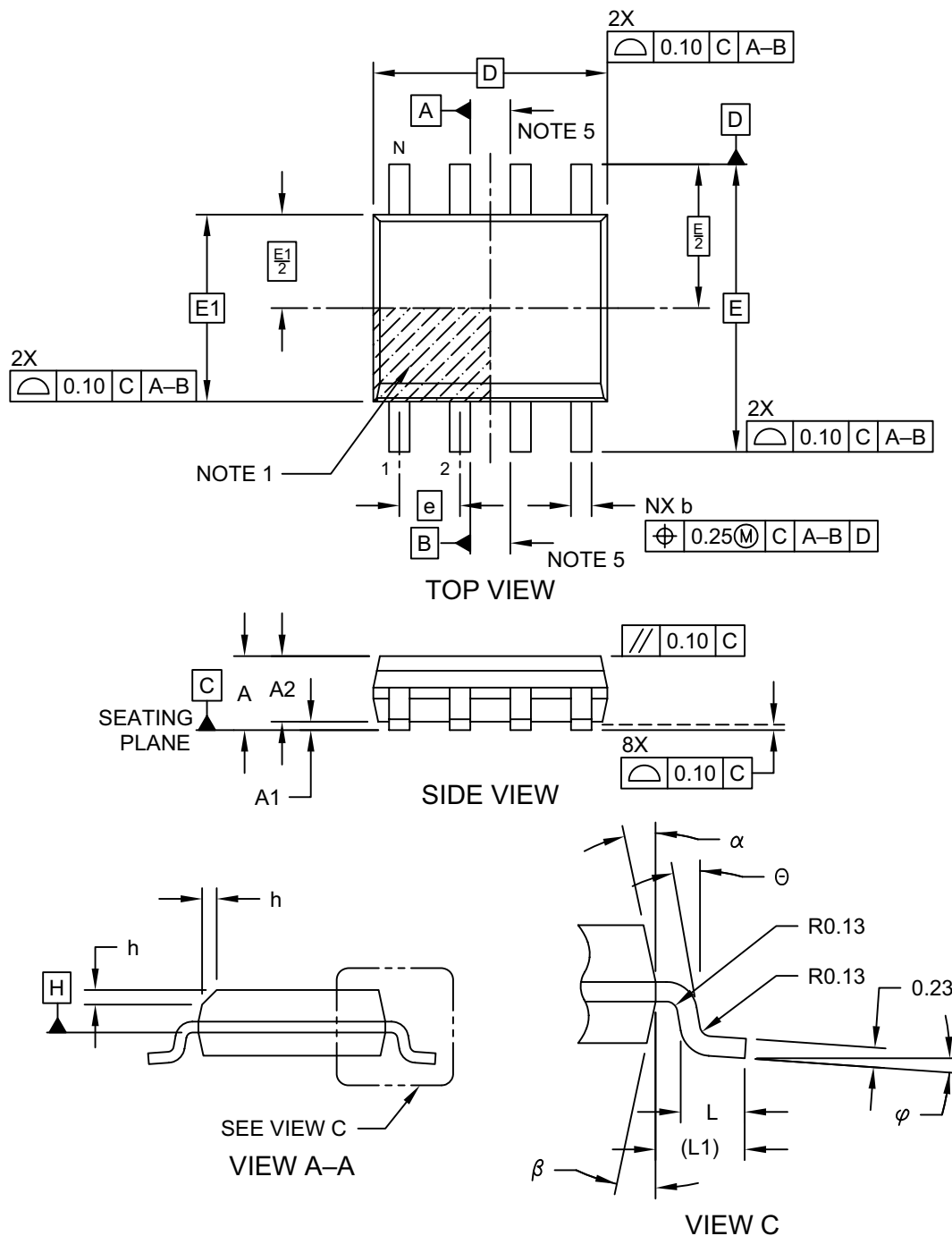
BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing No. C04-2062B

MCP2542FD/4FD, MCP2542WFD/4WFD

8-Lead Plastic Small Outline (SN) - Narrow, 3.90 mm (.150 In.) Body [SOIC]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>

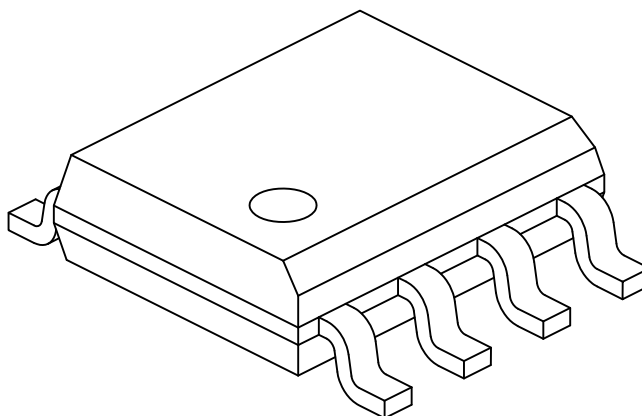


Microchip Technology Drawing No. C04-057-SN Rev F Sheet 1 of 2

MCP2542FD/4FD, MCP2542WFD/4WFD

8-Lead Plastic Small Outline (SN) - Narrow, 3.90 mm (.150 In.) Body [SOIC]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



		Units	MILLIMETERS		
Dimension Limits			MIN	NOM	MAX
Number of Pins	N		8		
Pitch	e		1.27 BSC		
Overall Height	A		-	-	1.75
Molded Package Thickness	A2		1.25	-	-
Standoff §	A1		0.10	-	0.25
Overall Width	E		6.00 BSC		
Molded Package Width	E1		3.90 BSC		
Overall Length	D		4.90 BSC		
Chamfer (Optional)	h		0.25	-	0.50
Foot Length	L		0.40	-	1.27
Footprint	L1		1.04 REF		
Foot Angle	φ		0°	-	8°
Lead Thickness	c		0.17	-	0.25
Lead Width	b		0.31	-	0.51
Mold Draft Angle Top	α		5°	-	15°
Mold Draft Angle Bottom	β		5°	-	15°

Notes:

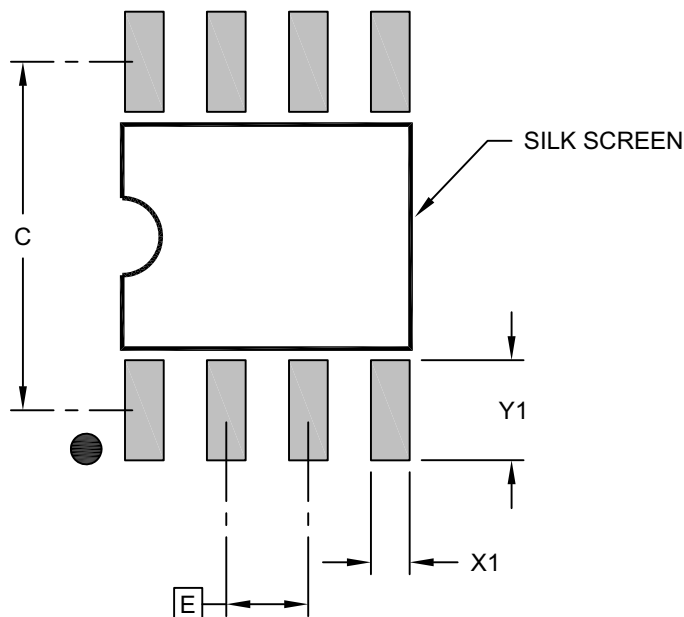
- Pin 1 visual index feature may vary, but must be located within the hatched area.
- § Significant Characteristic
- Dimensions D and E1 do not include mold flash or protrusions. Mold flash or protrusions shall not exceed 0.15mm per side.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.
- Datums A & B to be determined at Datum H.

Microchip Technology Drawing No. C04-057-SN Rev F Sheet 2 of 2

MCP2542FD/4FD, MCP2542WFD/4WFD

8-Lead Plastic Small Outline (SN) - Narrow, 3.90 mm (.150 In.) Body [SOIC]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Contact Pitch	E	1.27 BSC		
Contact Pad Spacing	C		5.40	
Contact Pad Width (X8)	X1			0.60
Contact Pad Length (X8)	Y1			1.55

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

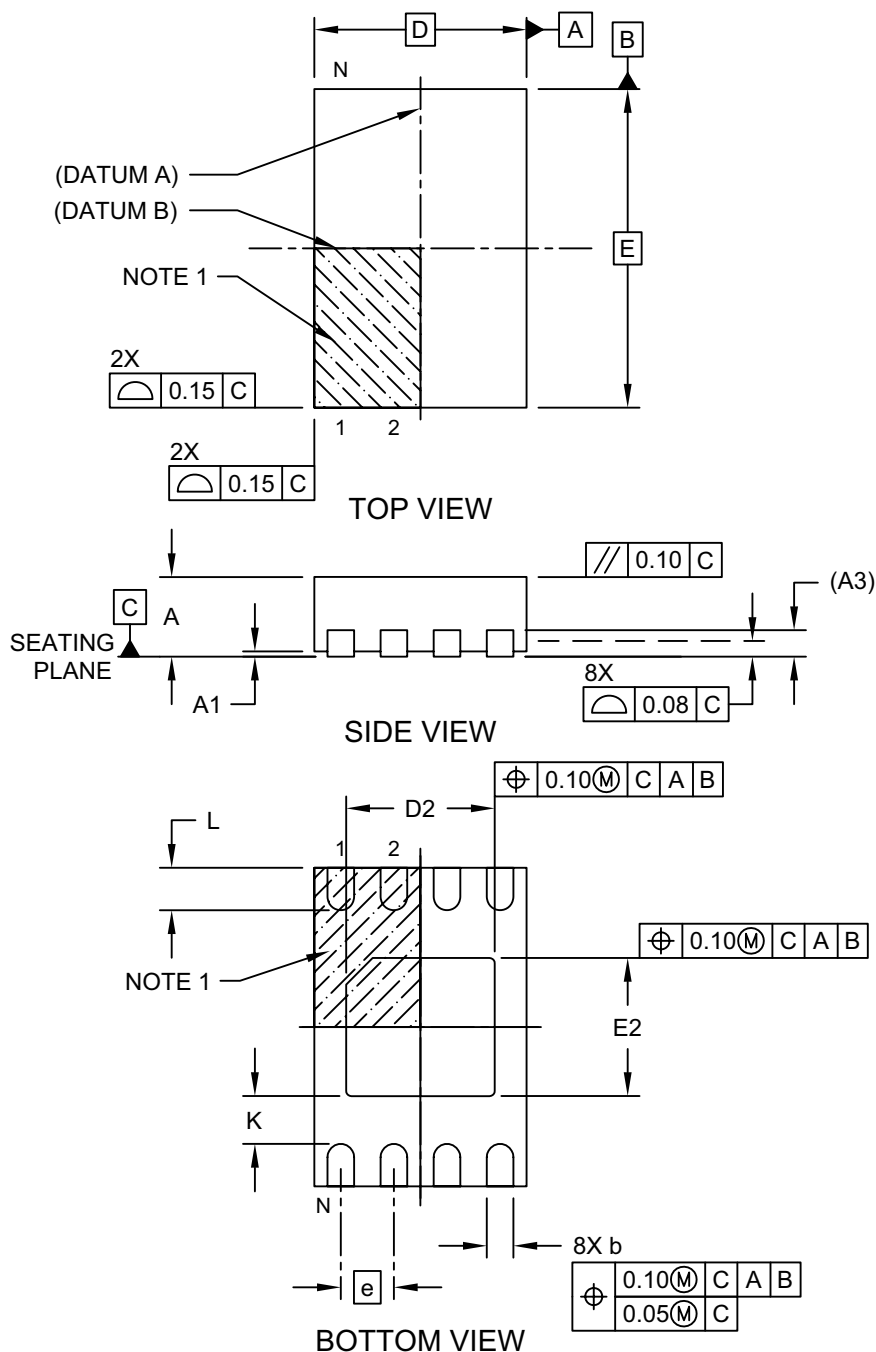
BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-2057-SN Rev F

MCP2542FD/4FD, MCP2542WFD/4WFD

8-Lead Plastic Dual Flat, No Lead Package (MNY) – 2x3x0.8 mm Body [TDFN] With 1.4x1.3 mm Exposed Pad (JEDEC Package type WDFN)

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>

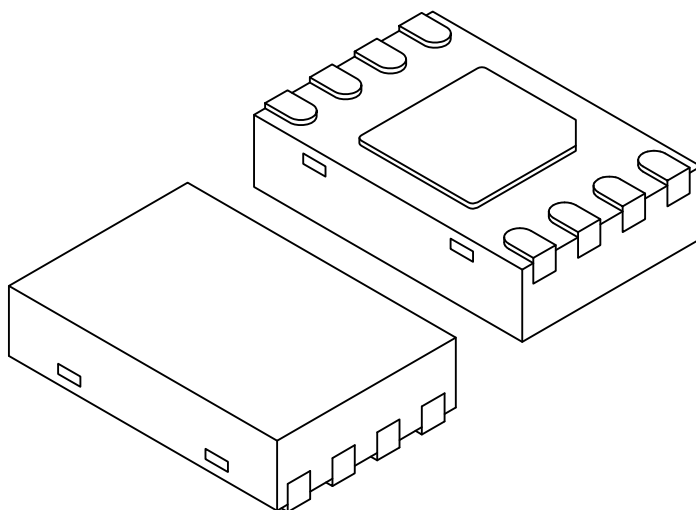


Microchip Technology Drawing No. C04-129-MNY Rev E Sheet 1 of 2

MCP2542FD/4FD, MCP2542WFD/4WFD

8-Lead Plastic Dual Flat, No Lead Package (MNY) – 2x3x0.8 mm Body [TDFN] With 1.4x1.3 mm Exposed Pad (JEDEC Package type WDFN)

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



		Units	MILLIMETERS		
Dimension Limits			MIN	NOM	MAX
Number of Pins	N		8		
Pitch	e		0.50 BSC		
Overall Height	A		0.70	0.75	0.80
Standoff	A1		0.00	0.02	0.05
Contact Thickness	A3		0.20 REF		
Overall Length	D		2.00 BSC		
Overall Width	E		3.00 BSC		
Exposed Pad Length	D2		1.35	1.40	1.45
Exposed Pad Width	E2		1.25	1.30	1.35
Contact Width	b		0.20	0.25	0.30
Contact Length	L		0.25	0.30	0.45
Contact-to-Exposed Pad	K		0.20	-	-

Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Package may have one or more exposed tie bars at ends.
3. Package is saw singulated
4. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

REF: Reference Dimension, usually without tolerance, for information purposes only.

Microchip Technology Drawing No. C04-129-MNY Rev E Sheet 2 of 2

MCP2542FD/4FD, MCP2542WFD/4WFD

NOTES:

MCP2542FD/4FD, MCP2542WFD/4WFD

APPENDIX A: REVISION HISTORY

Revision C (August 2020)

The following is the list of modifications:

- Updated **“Features”** section.
- Updated **Section 1.0 “Device Overview”**.
- Updated **Section 4.0 “Packaging Information”**.
- Updated **Section “Product Identification System”**.

Revision B (March 2019)

The following is the list of modifications:

- Changed High-Level Input Voltage for MCP2544FD and MCP2544WFD from VIO-0.3 to VDD-0.3 in **TABLE 2-1: “DC Characteristics”**.
- Fixed SOIC package markings in **Section 4.1 “Package Marking Information”**.
- Clarified that MCP2544FD/4FD is a CAN FD Transceiver without WUP Option in **Section “Product Identification System”**.
- Minor typographical corrections.

Revision A (February 2016)

Initial release of this document.

MCP2542FD/4FD, MCP2542WFD/4WFD

NOTES:

MCP2542FD/4FD, MCP2542WFD/4WFD

PRODUCT IDENTIFICATION SYSTEM

To order or obtain information, e.g., on pricing or delivery, refer to the factory or the listed sales office.

<u>PART NO.</u>	<u>[X]⁽¹⁾</u>	<u>X</u>	<u>/XX</u>
Device	Tape and Reel Option	Temperature Range	Package
Device: MCP2542FD/4FD: CAN FD Transceiver without WUP Option MCP2542WFD/4WFD: CAN FD Transceiver with WUP Option			
Tape and Reel Option: Blank = Standard packaging (tube or tray) T = Tape and Reel ⁽¹⁾			
Temperature Range: E = -40°C to +125°C (Extended) H = -40°C to +150°C (High)			
Package: MF = Plastic Dual Flat No Lead Package – 3 x 3 x 0.9 mm Body (DFN), 8-Lead MNY = Plastic Dual Flat No Lead Package – 2 x 3 x 0.8 mm Body (TDFN), 8-Lead SN = Plastic Small Outline – Narrow, 3.90 mm Body (SOIC), 8-Lead MFVAO = Plastic Dual Flat No Lead Package – 3 x 3 x 0.9 mm Body (DFN), 8-Lead, Automotive, AEC-Q100 Qualified MNYVAO = Plastic Dual Flat No Lead Package – 2 x 3 x 0.8 mm Body (TDFN), 8-Lead, Automotive, AEC-Q100 Qualified SNVAO = Plastic Small Outline – Narrow, 3.90 mm, Body (SOIC), 8-Lead, Automotive, AEC-Q100 Qualified			

Examples:
a) MCP2542FD-E/MF: Extended Temperature, 8-Lead, Plastic Dual Flat No Lead DFN package.
b) MCP2544WFD-H/MF: High Temperature, 8-Lead, Plastic Dual Flat No Lead DFN package.
c) MCP2542WFDT-H/SN: Tape and Reel, High Temperature, 8-Lead, Plastic Small Outline SOIC package.
d) MCP2544WFDT-E/SN: Tape and Reel, Extended Temperature, 8-Lead, Plastic Small Outline SOIC package.
e) MCP2542FDT-E/MNY: Tape and Reel, Extended Temperature, 8-Lead, Plastic Dual Flat No Lead TDFN package.
f) MCP2544WFDT-H/MNY: Tape and Reel, High Temperature, 8-Lead, Plastic Dual Flat No Lead TDFN package.
g) MCP2542FDT-E/MFVAO: Tape and Reel, Extended Temperature, 8-Lead, Plastic Dual Flat No Lead DFN package, Automotive AEC-Q100 Qualified.

Note 1: Tape and Reel identifier only appears in the catalog part number description. This identifier is used for ordering purposes and is not printed on the device package. Check with your Microchip Sales Office for package availability with the Tape and Reel option.

MCP2542FD/4FD, MCP2542WFD/4WFD

NOTES:

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- Microchip is willing to work with the customer who is concerned about the integrity of their code.
- Neither Microchip nor any other semiconductor manufacturer can guarantee the security of their code. Code protection does not mean that we are guaranteeing the product as "unbreakable."

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